# Predicting unplanned readmission for decompensation within one year in hepatitis B related decompensated cirrhosis

## Online Resource 1

Hepatology International

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Supplementary methods, extended results, Tables S1–S9, and Figs. S1–S2. References use the numbering in the main article.

## Note M1 Variable definitions coding and measurement

Tables S1 and S2 describe the Chinese and external cohorts. Table S3A lists 50 recorded characteristics and identifies the 46 predefined candidate predictors, with their definitions, units, analytical roles, and availability. Hepatic encephalopathy, prothrombin time, prothrombin activity, and MELD-Na were descriptive or benchmark variables. Obesity (BMI ≥28 kg/m²), hypertension, and diabetes mellitus were descriptive components of the metabolic-comorbidity composite and were not separate feature-selection candidates.

Baseline descriptions of the Chinese cohort use pre-imputation records and variable-specific nonmissing denominators. For HBV DNA, 205 results were quantitatively measured and 232 were left-censored with numerical bounds of 20, 50, 100, or 1,000 IU/mL. Half the recorded bound was substituted before $\log_{10}$ transformation. Five further results recorded as not detected without a numerical bound had no continuous value but were assigned the lowest category; three had no recorded result. The continuous and categorical denominators were therefore 437 and 442. Left-censoring was distinguished from ordinary missingness.

HBsAg was analyzed on the $\log_{10}$ IU/mL scale. The historical analytical convention used 0.05 and replaced recorded zero values with 0.025. This convention does not identify a verified universal assay detection limit or imply uniform treatment of all results below 0.05.

Multiple imputation in the Chinese cohort was performed after the development–validation split using development-estimated parameters. Five completed datasets were generated, and one fixed completed dataset was used for modeling. The retained historical imputation record identifies the second completed dataset and predictive mean matching for neutrophil, lymphocyte and monocyte counts and continuous HBV DNA. Validation data did not estimate imputation parameters; outcomes were neither imputed nor included as predictors of missing values. Derived indices used the formulas and rounding rules in Table S3A. Supplementary descriptive summaries use the original pre-imputation data.

External evaluation comprised 102 eligible hospitalization episodes from 50 patients, including 67 event and 35 non-event episodes. Inputs were observed measurements from the same continuous emergency–inpatient encounter, selected through an admission-oriented hierarchy without statistical imputation. Of 816 component selections, 804 fell within −24 to +24 hours of admission, nine after 24 and within 48 hours, and three followed the predefined extended hierarchy beyond 48 hours. Results had to be available by the applicable deadline and index discharge. Thus, measurements were closely aligned to admission but were not universally restricted to 48 hours. Table S2 specifies episode-level, patient-level, and variable-specific denominators. SMDs describe case mix. Clinical review and database-based postdischarge ascertainment are detailed in Note M6.

## Note M2 Fixed models nested evaluation and patient aggregation

LASSO at $\lambda_{1se}$ and Boruta with TentativeRoughFix independently evaluated all 46 predefined candidates in the development cohort. Their intersection defined Core4: INR, PLT, ALBI, and FIB-4. Parallel univariable logistic analyses did not filter either selection input. Spearman correlations and VIFs were assessed after selection (Table S3B and C). Table S4 gives final model settings and search configurations; final fixed hyperparameters do not specify those selected in every nested partition.

The repeated nested evaluation comprised 100 outcome-stratified partitions, each with 312 outer training and 133 outer test patients. Within each partition, chained-equation imputation generated five datasets over 20 iterations; the second completed dataset was used for selection and fitting. The imputation method and predictor matrix were initialized from outer training data, and outer test records were excluded from parameter estimation using the ignore setting. Outcomes were neither imputed nor used as imputation predictors. Missing HBV DNA, neutrophil, lymphocyte, and monocyte values used predictive mean matching; derived indices were recalculated afterward. LASSO and Boruta selection, tuning, and fitting then used outer training data only. All 100 selection procedures completed. The primary seven-model comparison used 98 compatible partitions; LASSO and elastic net had 95 valid evaluations because three additional compatible partitions yielded a single predictor unsupported by their fitting interface. Failed or excluded fits were not counted as valid results. Table S5A gives medians and quartiles across the actual valid partitions, which describe resampling variability rather than CIs.

For nested LASSO selection, continuous candidates were min–max scaled using outer training values; a binomial glmnet model used tenfold cross-validation, 1,000 candidate penalty values, standardize = false, and $\lambda_{1se}$. Boruta used pValue = 0.01, multiplicity adjustment, maxRuns = 1,000, and random-forest Z-score importance, followed by TentativeRoughFix where needed. These settings describe the nested selection procedure; Table S4 separately records the fixed development-model settings.

For the nested RBF-SVM–MELD-Na comparison, 98 paired outer test partitions contributed predictions. Each patient's risk was the mean of predictions obtained only when that patient was held out. All 445 Chinese patients (185 events) were represented, with 15–45 held-out predictions each (median, 29). The MELD-Na logistic benchmark was fitted within the corresponding outer training partitions. Table S5D presents this patient-level aggregate separately from partition-specific distributions. Paired patient-bootstrap intervals condition on the aggregated predictions; model development was not repeated within bootstrap samples. The aggregate remains an internal evaluation of the Chinese cohort.

## Note M3 Predictor representations ablations and event timing

Table S6A compares three representations using logistic regression and RBF-SVM in both fixed evaluation cohorts. Original Core4 comprised INR, PLT, ALBI, and FIB-4. The raw-variable candidate pool retained BMI and excluded length of stay; its selected intersection was INR and PLT. The composite-variable pool retained 11 derived indices and excluded their ten raw components, leaving 36 candidates; its intersection was PTAR and FIB-4. Length of stay remained a candidate in the composite analysis but was not selected. The composite pool therefore included a variable available only after admission. AUROC, AP, Brier score, CITL, and calibration slope are reported as point estimates.

Table S6B evaluates six ablations: E1 omitted INR; E2 omitted PLT; E3 omitted ALBI; E4 omitted FIB-4; E5 retained INR and ALBI; and E6 retained PLT and FIB-4. $\Delta$AUROC and $\Delta$AP are the full model minus the ablated model, whereas Brier improvement is the ablated model minus the full model. Positive differences favor the full model. Reduced models were fitted in the original 312-patient development cohort without repeating feature selection; full-model predictions were unchanged.

CIs were obtained from paired differences, rather than subtraction of marginal intervals, and condition on the evaluated predictions without repeating model development. These exploratory comparisons were unadjusted for multiplicity. Intervals crossing zero do not establish equivalence or absence of a predictor contribution; ablations evaluate prediction rather than independent causal effects.

Table S7 evaluates scores from the six original fixed 1-year models, without fitting separate early or late models. Early events occurred at 0–90 days, and later events after 90 and within 365 days. Each event group was compared with the same cohort-specific 1-year non-event controls; the other event group was excluded from that comparison. Internal counts were 22 early events, 33 later events, and 78 shared controls. External counts were 48 early events, 17 later events, and 35 shared controls. Two external event episodes with unclassifiable timing remained in the annual analysis but were excluded from both timing comparisons.

The external early comparison included 83 episodes from 48 patients; the later comparison included 52 episodes from 37 patients. These overlapping sets cannot be summed to reconstruct the primary cohort. The 5,000 full-cohort bootstrap samples resampled patients internally and patient clusters externally, with early and later contrasts evaluated in the same resamples. CIs and later-minus-earlier differences retain their paired definitions. These analyses assess event-time-stratified discrimination of annual scores; they are not cumulative 90-day ROC analyses or estimates of time-specific calibrated probabilities.

## Note M4 Clinical benchmarking calibration and decision measures

The fixed MELD-Na benchmark was a development-fitted single-predictor logistic model applied unchanged to both evaluation cohorts. Table S5B reports absolute performance and diagnostic calibration; Table S5C reports fixed paired comparisons. Calibration fits were diagnostic and did not replace predictions. Higher AUROC and AP and lower Brier scores indicate better performance. Ideal CITL is zero; a positive CITL indicates overall underprediction. Ideal calibration slope and observed-to-expected ratio (O/E) are one. AP is not positive predictive value at a single threshold.

Fig. 4d and e group observations separately for each model using the original evaluation sample's type 7 prediction quantiles: five groups internally and four externally. Duplicate boundaries are merged, and equal predictions are not split across groups. Points show group mean predicted probabilities and observed event proportions. Group membership remains fixed during resampling. Vertical error bars show 95% percentile bootstrap intervals for group observed proportions, using 5,000 patient resamples internally and patient-cluster resamples externally; they are distinct from the Wilson intervals in Fig. 4c. Horizontal error bars are not plotted.

Calibration curves are diagnostic logistic fits of the observed outcome on the logit of the original predicted probability, with a freely estimated intercept and slope. Curves connect saved coordinates only within each model's observed probability range; curve confidence bands are not displayed. The free intercept used for the curve differs from CITL, which estimates an intercept with the logit prediction as an offset and its coefficient fixed at one. Probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$ only for logit-based diagnostic calculations. These fits characterize calibration and do not recalibrate or replace the predictions used for performance assessment.

Fig. 4c and Table S9 use tertiles of the 312 development-cohort RBF-SVM predictions, calculated using type 7 quantiles without optimizing outcome separation. The fixed cutoffs are 0.234116093557522 and 0.559109361430369: low risk, $p \leq$ the lower cutoff; intermediate risk, lower cutoff $< p \leq$ upper cutoff; high risk, $p >$ the upper cutoff. Event-rate intervals are Wilson intervals internally and patient-cluster bootstrap intervals externally. Coverage is the fraction of eligible observations selected; event capture is the fraction of all observed events among selected observations; missed events occur outside that group. These cutoffs describe the development risk distribution and do not define validated treatment or follow-up thresholds.

Fig. S1 compares three models at nominal capacities of 10%, 20%, 30%, 40%, and 50%. Each strategy selected $K = \text{floor}(qN)$ observations, where q is nominal capacity and N is the record count; actual coverage K/N can therefore differ from q. Ranking ties followed a fixed outcome-blind rule. Pointwise intervals used 5,000 resamples, repeating selection within each resample. Resampling used patients internally and patient clusters retaining all eligible episodes externally. Intervals condition on the fixed predictions.

Fig. S2 shows decision thresholds from 0.05 to 0.80, including treat-all, treat-none, negative net benefit, and intervals crossing zero. Paired differences are RBF-SVM minus either MELD-Na or Core4 logistic

regression. Models shared the same bootstrap samples within each cohort, with patient clustering externally; intervals were calculated from within-sample contrasts. Intervals are pointwise rather than simultaneous across thresholds. Neither an optimal threshold nor an optimal capacity was selected. Positive estimated net benefit concerns a hypothetical action and requires prospective evaluation before it can imply clinical benefit.

## Note M5 Prediction attribution and reference background

Exact Kernel SHAP used kernelshap version 0.9.1 on the positive-class probability scale. The fixed RBF-SVM and its prediction wrapper were retained. For each explained record, all four-feature coalitions were evaluated against the same empirical background of 312 locked development records, without coalition sampling. The common baseline model output was 0.416286863888495. The background matrix differed from the original fitted-model input in one recorded FIB-4 value, which was 7.12 in the reference matrix and 7.13 in the original training matrix; the model was not refitted and neither matrix was altered for the attribution calculation. The background therefore represents the locked development reference rather than a byte-identical copy of the original fit matrix.

Saved attribution checks confirmed agreement with the fixed-model predictions and local additivity. The maximum absolute error in baseline + ΣSHAP − prediction was $1.11022302462516 \times 10^{-16}$ in both evaluation cohorts. An independent explicit-coalition implementation checked three internal records selected solely by proximity to the 10th, 50th, and 90th percentiles of predicted risk; the current external calculations also underwent an independent direct check. These numerical checks are distinct from the two illustrative cases displayed in Fig. 5g and h, which were selected after inspection of predictions and outcomes.

The attributions are exact relative to the specified empirical reference and probability function, rather than exact biological effects. Background-based feature replacement can produce combinations outside observed physiological relationships, particularly because PLT is a component of FIB-4. Attribution patterns should be interpreted jointly and do not establish causality or monotonicity over all possible predictor values.

## Table S1 Outcome-stratified characteristics of the Chinese cohort

### A Development cohort

| Characteristic | No target readmission (n = 182) | Target readmission (n = 130) | P value |
|---|---|---|---|
| Demographic and background characteristics | | | |
| Age (years) | 51.0 (44.0, 60.0) | 55.0 (47.0, 63.0) | 0.029 |
| Male sex | 132 (72.5) | 87 (66.9) | 0.286 |
| Body mass index (kg/m²) | 22.42 (20.32, 24.45) | 22.59 (20.99, 24.43) | 0.440 |
| Metabolic comorbidities | 55 (30.2) | 38 (29.2) | 0.851 |
| Smoking history | 78 (42.9) | 54 (41.5) | 0.816 |
| Alcohol use history | 77 (42.3) | 49 (37.7) | 0.413 |
| HBV-related virological characteristics | | | |
| HBeAg/anti-HBe status | | | 0.971 |
| HBeAg-/anti-HBe+ | 127 (69.8) | 88 (67.7) | |
| HBeAg+/anti-HBe- | 32 (17.6) | 24 (18.5) | |
| HBeAg-/anti-HBe- | 15 (8.2) | 11 (8.5) | |
| HBeAg+/anti-HBe+ | 8 (4.4) | 7 (5.4) | |
| HBV DNA study category (IU/mL) | n = 182 | n = 127 | 0.001 |
| <2,000 | 120 (65.9) | 59 (46.5) | |
| 2,000-<20,000 | 17 (9.3) | 12 (9.4) | |
| ≥20,000 | 45 (24.7) | 56 (44.1) | |
| HBV DNA, log10 (IU/mL) | 2.70 (2.70, 4.38); n = 178 | 3.81 (2.70, 5.83); n = 126 | 0.021 |
| HBsAg, log10 (IU/mL) | 2.76 (1.90, 3.09) | 2.87 (2.17, 3.16) | 0.086 |
| Prior antiviral therapy | 105 (57.7) | 73 (56.2) | 0.787 |
| Recorded clinical manifestations | | | |
| Ascites | 69 (37.9) | 61 (46.9) | 0.111 |
| EGVH | 109 (59.9) | 64 (49.2) | 0.062 |
| SBP | 37 (20.3) | 28 (21.5) | 0.795 |
| HE | 4 (2.2) | 19 (14.6) | <0.001 |
| HRS | 1 (0.5) | 1 (0.8) | 1.000 |
| Infection | 66 (36.3) | 56 (43.1) | 0.224 |

| Characteristic | No target readmission (n = 182) | Target readmission (n = 130) | P value |
|---|---|---|---|
| Laboratory measurements | | | |
| Albumin (g/L) | 31.00 (27.00, 34.00) | 28.00 (25.00, 32.00) | <0.001 |
| Total bilirubin (μmol/L) | 20.05 (13.93, 31.18) | 34.15 (22.15, 50.30) | <0.001 |
| ALT (U/L) | 36.00 (26.00, 51.75) | 45.00 (31.00, 71.50) | 0.004 |
| AST (U/L) | 40.00 (29.00, 55.75) | 49.00 (33.25, 98.00) | 0.001 |
| ALP (U/L) | 91.00 (68.00, 118.50) | 96.00 (68.00, 143.50) | 0.110 |
| GGT (U/L) | 34.00 (23.25, 64.50) | 40.00 (23.00, 80.50) | 0.253 |
| INR | 1.28 (1.18, 1.39) | 1.48 (1.37, 1.63) | <0.001 |
| PT (s) | 15.60 (14.50, 17.00) | 17.80 (16.12, 19.38) | <0.001 |
| PTA (%) | 66.00 (58.02, 75.80) | 53.30 (46.00, 59.00) | <0.001 |
| Platelet count (10^9/L) | 77.00 (54.00, 132.00) | 51.00 (37.00, 72.50) | <0.001 |
| WBC (10^9/L) | 4.25 (2.85, 6.89) | 3.43 (2.48, 5.26) | 0.003 |
| Hemoglobin (g/L) | 89.00 (70.25, 108.00) | 89.50 (72.00, 111.00) | 0.496 |
| Serum creatinine (μmol/L) | 68.00 (56.25, 79.00) | 64.00 (55.00, 77.75) | 0.273 |
| Sodium (mmol/L) | 139.00 (137.00, 141.00) | 139.00 (137.00, 142.00) | 0.927 |
| Composite scores | | | |
| MELD-Na | 10.87 (9.20, 13.49) | 14.19 (11.83, 17.20) | <0.001 |
| ALBI | -1.81 (-2.10, -1.39) | -1.38 (-1.77, -1.04) | <0.001 |
| FIB-4 | 4.38 (2.70, 7.01) | 9.60 (5.06, 13.91) | <0.001 |
| Index hospitalization | | | |
| Length of index hospital stay (days) | 10.0 (7.0, 14.0) | 11.0 (7.0, 15.0) | 0.451 |

## B Fixed internal validation cohort

| Characteristic | No target readmission (n = 78) | Target readmission (n = 55) | P value |
|---|---|---|---|
| Demographic and background characteristics | | | |
| Age (years) | 53.5 (47.0, 59.0) | 50.0 (45.0, 60.0) | 0.261 |
| Male sex | 55 (70.5) | 39 (70.9) | 0.961 |
| Body mass index (kg/m²) | 21.86 (20.31, 23.22) | 23.39 (21.50, 25.52) | 0.003 |
| Metabolic comorbidities | 23 (29.5) | 15 (27.3) | 0.781 |
| Smoking history | 35 (44.9) | 18 (32.7) | 0.159 |
| Alcohol use history | 35 (44.9) | 22 (40.0) | 0.576 |

| Characteristic | No target readmission (n = 78) | Target readmission (n = 55) | P value |
|---|---|---|---|
| HBV-related virological characteristics | | | |
| HBeAg/anti-HBe status | | | 0.461 |
| HBeAg-/anti-HBe+ | 56 (71.8) | 42 (76.4) | |
| HBeAg+/anti-HBe- | 11 (14.1) | 10 (18.2) | |
| HBeAg-/anti-HBe- | 8 (10.3) | 2 (3.6) | |
| HBeAg+/anti-HBe+ | 3 (3.8) | 1 (1.8) | |
| HBV DNA study category (IU/mL) | | | 0.623 |
| <2,000 | 48 (61.5) | 33 (60.0) | |
| 2,000-<20,000 | 12 (15.4) | 6 (10.9) | |
| ≥20,000 | 18 (23.1) | 16 (29.1) | |
| HBV DNA, log10 (IU/mL) | 2.70 (2.70, 4.16) | 2.70 (2.70, 4.58) | 0.357 |
| HBsAg, log10 (IU/mL) | 2.71 (1.88, 3.14) | 2.91 (2.01, 3.28) | 0.139 |
| Prior antiviral therapy | 39 (50.0) | 29 (52.7) | 0.757 |
| Recorded clinical manifestations | | | |
| Ascites | 40 (51.3) | 24 (43.6) | 0.385 |
| EGVH | 42 (53.8) | 31 (56.4) | 0.774 |
| SBP | 23 (29.5) | 20 (36.4) | 0.404 |
| HE | 1 (1.3) | 3 (5.5) | 0.306 |
| HRS | 3 (3.8) | 0 (0.0) | 0.267 |
| Infection | 33 (42.3) | 26 (47.3) | 0.570 |
| Laboratory measurements | | | |
| Albumin (g/L) | 31.00 (27.00, 36.00) | 27.00 (25.00, 32.50) | 0.002 |
| Total bilirubin (μmol/L) | 21.60 (13.75, 35.42) | 32.80 (18.05, 49.20) | 0.005 |
| ALT (U/L) | 38.50 (27.25, 51.75) | 41.00 (28.00, 56.50) | 0.819 |
| AST (U/L) | 42.50 (31.25, 62.25) | 45.00 (30.00, 74.00) | 0.766 |
| ALP (U/L) | 92.50 (67.00, 129.00) | 90.00 (65.50, 137.00) | 0.980 |
| GGT (U/L) | 55.00 (27.25, 108.75) | 40.00 (23.00, 65.50) | 0.069 |
| INR | 1.26 (1.19, 1.41) | 1.52 (1.35, 1.71) | <0.001 |
| PT (s) | 15.75 (15.03, 17.20) | 18.30 (16.00, 20.05) | <0.001 |
| PTA (%) | 66.40 (57.15, 75.00) | 53.00 (44.50, 62.30) | <0.001 |

| Characteristic | No target readmission (n = 78) | Target readmission (n = 55) | P value |
|---|---|---|---|
| Platelet count (10^9/L) | 67.00 (54.25, 94.75) | 51.00 (36.00, 66.50) | 0.001 |
| WBC (10^9/L) | 4.07 (2.78, 5.64) | 4.06 (2.87, 5.54) | 0.866 |
| Hemoglobin (g/L) | 97.50 (78.25, 115.75) | 95.00 (79.00, 108.00) | 0.264 |
| Serum creatinine (μmol/L) | 71.00 (60.25, 83.75) | 66.00 (58.00, 74.50) | 0.038 |
| Sodium (mmol/L) | 138.00 (136.00, 140.75) | 138.00 (136.00, 141.00) | 0.927 |
| Composite scores | | | |
| MELD-Na | 11.71 (9.29, 14.74) | 14.33 (12.25, 17.57) | <0.001 |
| ALBI | -1.73 (-2.17, -1.39) | -1.36 (-1.74, -1.03) | <0.001 |
| FIB-4 | 5.04 (2.88, 7.99) | 7.26 (5.07, 11.15) | 0.005 |
| Index hospitalization | | | |
| Length of index hospital stay (days) | 11.5 (8.0, 19.0) | 10.0 (8.0, 14.0) | 0.371 |

Values are medians (Q1, Q3) or n (%) from available pre-imputation records, with percentages based on nonmissing denominators. Available HBV DNA n in the development non-event/event groups was 178/126 for the continuous measure and 182/127 for the categorical measure. All displayed analytical fields were available in both fixed internal-validation groups.

P values compare outcome groups within each cohort using two-sided Mann–Whitney U tests for continuous variables and chi-squared or Fisher exact tests for categorical variables. Comparisons were exploratory and unadjusted for multiplicity and did not determine entry into feature selection. Clinical manifestations can coexist.

Metabolic comorbidity denotes obesity, hypertension, or diabetes mellitus; obesity was defined as BMI ≥28 kg/m². HBeAg/anti-HBe status represents the four combinations of positive and negative results. Viral-marker coding and measurement limitations are detailed in Table S3A.

Abbreviations: HBV, hepatitis B virus; DNA, deoxyribonucleic acid; HBeAg, hepatitis B e antigen; anti-HBe, antibody to hepatitis B e antigen; HBsAg, hepatitis B surface antigen; EGVH, esophagogastric variceal hemorrhage; SBP, spontaneous bacterial peritonitis; HE, hepatic encephalopathy; HRS, hepatorenal syndrome; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ALP, alkaline phosphatase; GGT, gamma-glutamyl transferase; INR, international normalized ratio; PT, prothrombin time; PTA, prothrombin activity; WBC, white blood cell count; MELD-Na, Model for End-Stage Liver Disease-sodium; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index; Q1/Q3, first/third quartiles.

## Table S2 External cohort characteristics and case-mix comparison

### A External cohort structure and index-hospitalization profile

| Characteristic | External cohort |
|---|---|
| Cohort structure | |
| Eligible index hospitalization episodes | 102 |
| Unique patients | 50 |
| Eligible episodes per patient | 1.00 [1.00, 2.75] |
| Patients contributing 1 eligible episode | 34 (68.0%) |
| Patients contributing 2 eligible episodes | 3 (6.0%) |
| Patients contributing ≥3 eligible episodes | 13 (26.0%) |
| 1-year readmission events | 67/102 (65.7%) |
| Demographics | |
| Age, years | 58.00 [54.00, 65.00] |
| Male sex | 86/102 (84.3%) |
| Race/ethnicity, n (%) | |
| Asian | 26 (25.5%) |
| Black | 12 (11.8%) |
| Hispanic/Latino | 10 (9.8%) |
| White | 52 (51.0%) |
| Other/unknown | 2 (2.0%) |
| Index hospitalization characteristics | |
| Hospital length of stay, days | 4.16 [2.55, 6.89] |
| Any ICU admission during index hospitalization | 13/102 (12.7%) |
| ICU length of stay among ICU-exposed episodes, days | 1.80 [0.89, 2.10]; n = 13 |
| Index decompensation manifestations, n/N (%) | |
| Ascites | 84/101 (83.2%) |
| Hepatic encephalopathy | 26/101 (25.7%) |
| Variceal bleeding | 7/101 (6.9%) |
| Spontaneous bacterial peritonitis | 8/101 (7.9%) |
| Hepatorenal syndrome | 2/101 (2.0%) |

## B Chinese and external case-mix comparison

| Characteristic | Chinese cohort (N = 445 patients) | External (N = 102 episodes) | SMD |
|---|---|---|---|
| Demographics | | | |
| Age, years | 52.00 [45.00, 61.00] | 58.00 [54.00, 65.00] | -0.582 |
| Male sex | 313/445 (70.3%) | 86/102 (84.3%) | -0.339 |
| Liver and laboratory profile | | | |
| Albumin, g/L | 29.00 [26.00, 34.00] | 29.00 [26.00, 34.00] | 0.036 |
| Total bilirubin, μmol/L | 24.70 [15.70, 41.00] | 35.91 [17.10, 63.27] | -0.411 |
| INR | 1.37 [1.22, 1.52] | 1.50 [1.40, 1.70] | -0.504 |
| Platelet count, ×10^9/L | 63.00 [44.00, 95.00] | 67.50 [42.50, 101.75] | -0.076 |
| Creatinine used for MELD-Na, μmol/L | 67.00 [56.00, 79.00] | 88.40 [70.72, 112.71] | -0.606 |
| Sodium used for MELD-Na, mmol/L | 139.00 [137.00, 141.00] | 135.00 [131.00, 138.75] | 0.921 |
| Composite liver-disease indices | | | |
| ALBI score | -1.62 [-2.02, -1.22] | -1.49 [-1.95, -1.09] | -0.165 |
| FIB-4 | 5.76 [3.38, 10.53] | 9.65 [5.60, 15.39] | -0.484 |
| MELD-Na | 12.47 [9.99, 15.54] | 18.43 [13.89, 23.69] | -1.044 |

Values are medians (Q1, Q3) or n/N (%) unless otherwise stated. Hospitalization episodes are the analysis unit, and patients are the clustering unit for uncertainty estimation. Patient-contribution percentages use 50 patients; race/ethnicity percentages use 102 episodes. ICU length of stay is summarized across the 13 ICU-exposed episodes.

Index decompensation manifestations were identified through the MIMIC-specific structured-data adjudication. Manifestations can coexist; denominators reflect available phenotype-support information. Race/ethnicity categories were collapsed from recorded MIMIC-IV categories for description.

SMD is Chinese minus external using pooled standard deviations; positive values indicate higher Chinese-cohort values. SMDs describe case mix and were not used for hypothesis testing or model selection. Chinese-cohort measures reflected index-admission status; external measures were aligned to admission as closely as availability allowed. Case-mix comparisons do not isolate causes of performance differences.

The admission-oriented measurement hierarchy and its exceptions are described in Note M1.

Abbreviations: ICU, intensive care unit; MIMIC, Medical Information Mart for Intensive Care; SMD, standardized mean difference; INR, international normalized ratio; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index; MELD-Na, Model for End-Stage Liver Disease-sodium; Q1/Q3, first/third quartiles. Core4 comprises INR, platelet count, ALBI and FIB-4.

## Table S3 Predictor definitions availability and selection

### A Predictor definitions and availability profiles

| Variable | Unit/scale | Role | Definition/coding | Profile |
|---|---|---|---|---|
| Metabolic comorbidities | Category | Candidate | Obesity OR hypertension OR diabetes mellitus | A |
| EGVH | Category | Candidate | Stored definition | A |
| Ascites | Category | Candidate | Stored definition | A |
| SBP | Category | Candidate | Stored definition | A |
| HE | Category | Description only | Stored definition | A |
| HRS | Category | Candidate | Stored definition | A |
| Infection | Category | Candidate | Stored definition | A |
| HBeAg/anti-HBe status | Category | Candidate | Stored definition | A |
| Sex | Category | Candidate | Stored definition | A |
| Prior antiviral therapy | Category | Candidate | Stored definition | A |
| Smoking history | Category | Candidate | Stored definition | A |
| Alcohol use history | Category | Candidate | Stored definition | A |
| Serum creatinine (μmol/L) | μmol/L | Candidate | Stored definition | A |
| Total bilirubin (μmol/L) | μmol/L | Candidate | Stored definition | A |
| ALT (U/L) | U/L | Candidate | Stored definition | A |
| AST (U/L) | U/L | Candidate | Stored definition | A |
| GGT (U/L) | U/L | Candidate | Stored definition | A |
| LDH (U/L) | U/L | Candidate | Stored definition | A |
| ALP (U/L) | U/L | Candidate | Stored definition | A |
| Albumin (g/L) | g/L | Candidate | Stored definition | A |
| PTA (%) | % | Description only | Stored definition | A |
| PT (s) | s | Description only | Stored definition | A |
| Hemoglobin (g/L) | g/L | Candidate | Stored definition | A |
| NEU (10^9/L) | 10^9/L | Candidate | Stored definition | B |
| LYM (10^9/L) | 10^9/L | Candidate | Stored definition | B |
| MON (10^9/L) | 10^9/L | Candidate | Stored definition | B |
| Platelet count (10^9/L) | 10^9/L | Candidate; Core4 | Stored definition | A |
| WBC (10^9/L) | 10^9/L | Candidate | Stored definition | A |

| Variable | Unit/scale | Role | Definition/coding | Profile |
|---|---|---|---|---|
| Potassium (mmol/L) | mmol/L | Candidate | Stored definition | A |
| Sodium (mmol/L) | mmol/L | Candidate | Stored definition | A |
| Chloride (mmol/L) | mmol/L | Candidate | Stored definition | A |
| INR | ratio | Candidate; Core4 | Stored definition | A |
| HBsAg, log10 (IU/mL) | log10 (IU/mL) | Candidate | Viral-marker coding below | A |
| HBV DNA study category (IU/mL) | IU/mL cutpoints | Candidate | 0: <2,000; 1: 2,000-<20,000; 2: ≥20,000 IU/mL | C |
| HBV DNA, log10 (IU/mL) | log10 (IU/mL) | Candidate | Viral-marker coding below | D |
| Age (years) | years | Candidate | Stored definition | A |
| MELD-Na | score | Description only | Formula below | A |
| ALBI | score | Candidate; Core4 | Formula below | A |
| PMR | Formula-defined index/ratio | Candidate | Formula below | B |
| PTAR | Formula-defined index/ratio | Candidate | Formula below | A |
| NAR | Formula-defined index/ratio | Candidate | Formula below | B |
| NLR | Formula-defined index/ratio | Candidate | Formula below | B |
| FIB-4 | score | Candidate; Core4 | Formula below | A |
| LMR | Formula-defined index/ratio | Candidate | Formula below | B |
| PLR | Formula-defined index/ratio | Candidate | Formula below | B |
| SIRI | Formula-defined index/ratio | Candidate | Formula below | B |
| PNI | Formula-defined index/ratio | Candidate | Formula below | B |
| APRI | Formula-defined index/ratio | Candidate | Formula below | A |
| Length of index hospital stay (days) | days | Candidate | Stored definition | A |
| Body mass index | kg/m² | Candidate | Stored definition | A |

All 50 recorded characteristics are listed; 46 were candidate predictors. HE, PT, PTA, and MELD-Na were descriptive or benchmark variables. Core4 comprises INR, PLT, ALBI, and FIB-4. Non-viral continuous variables retain their recorded numerical scales. Clinical categories reflect recorded classifications; measurement times and treatment timing were not assumed to be uniform.

## A Pre-imputation availability by cohort

| Profile | Overall N = 445 | Development N = 312 | Internal N = 133 |
|---|---|---|---|
| A | 445 / 0 | 312 / 0 | 133 / 0 |
| B | 438 / 7 | 306 / 6 | 132 / 1 |
| C | 442 / 3 | 309 / 3 | 133 / 0 |
| D | 437 / 8 | 304 / 8 | 133 / 0 |

Cells show available n / missing n. The profile assigned to each variable specifies its pre-imputation counts. Numerically encoded left-censored results are not counted as ordinary missingness.

## A Pre-imputation availability by outcome group

| Profile | Development no readmission N = 182 | Development readmission N = 130 | Internal no readmission N = 78 | Internal readmission N = 55 |
|---|---|---|---|---|
| A | 182 / 0 | 130 / 0 | 78 / 0 | 55 / 0 |
| B | 178 / 4 | 128 / 2 | 77 / 1 | 55 / 0 |
| C | 182 / 0 | 127 / 3 | 78 / 0 | 55 / 0 |
| D | 178 / 4 | 126 / 4 | 78 / 0 | 55 / 0 |

Cells show available n / missing n before imputation. Completed modeling datasets constitute a separate data state.

## A Metabolic-comorbidity components

| Component | Unit/scale | Definition |
|---|---|---|
| Obesity | Binary (0/1) | Derived from BMI ≥28 kg/m². |
| Hypertension | Binary (0/1) | Recorded structured diagnosis flag. |
| Diabetes mellitus | Binary (0/1) | Recorded structured diagnosis flag. |

These variables describe components of the metabolic-comorbidity composite and were not separate feature-selection candidates.

## A Derived predictor definitions

| Predictor | Formula |
|---|---|
| MELD-Na | Scr_mg_dl = min(4, max(1, Scr/88.4)); TBIL_mg_dl = max(1, TBIL/17.1); M = 3.78 × ln(TBIL_mg_dl) + 11.2 × ln(max(1, INR)) + 9.57 × ln(Scr_mg_dl) + 6.43; Na = min(137, max(125, Sodium)); round(M + 1.32 × (137-Na) - 0.033 × M × (137-Na), 2). |
| ALBI | round(0.66 × log10(TBIL) - 0.085 × Albumin, 2) |
| PMR | PLT/MON (MON > 0); no rounding |
| PTAR | 10 × INR/Albumin (Albumin > 0); no rounding |
| NAR | 10 × NEU/Albumin (Albumin > 0); no rounding |
| NLR | round(NEU/LYM, 2) |
| FIB-4 | round(Age × AST / (PLT × sqrt(ALT)), 2) |
| LMR | round(LYM/MON, 2) |

| Predictor | Formula |
|---|---|
| PLR | round(PLT/LYM, 2) |
| SIRI | round(NEU × MON/LYM, 2) |
| PNI | round(Albumin + 5 × LYM, 2) |
| APRI | round((AST/40) × 100/PLT, 2) |

Albumin is in g/L, bilirubin and creatinine in μmol/L, and cell counts in $10^9$/L. The study-specific definition PTAR = 10 × INR/albumin is used.

## A Additional descriptions Overall

| Variable | Overall (N = 445) |
|---|---|
| LDH (U/L) | 388.00 (288.00, 520.00) |
| NEU (10^9/L) | 2.68 (1.75, 4.74); n=438 |
| LYM (10^9/L) | 0.69 (0.47, 1.02); n=438 |
| MON (10^9/L) | 0.31 (0.20, 0.46); n=438 |
| Potassium (mmol/L) | 3.80 (3.50, 4.20) |
| Chloride (mmol/L) | 107.00 (104.00, 109.00) |
| PMR | 221.72 (143.92, 322.14); n=438 |
| PTAR | 0.45 (0.38, 0.57) |
| NAR | 0.93 (0.57, 1.69); n=438 |
| NLR | 3.69 (2.50, 6.75); n=438 |
| LMR | 2.40 (1.67, 3.42); n=438 |
| PLR | 92.52 (63.41, 143.15); n=438 |
| SIRI | 1.08 (0.56, 2.64); n=438 |
| PNI | 34.02 (29.81, 37.85); n=438 |
| APRI | 1.73 (1.01, 3.27) |

Values are medians (Q1, Q3), with available n when it differs from the group total. Outcome-group P values are exploratory, unadjusted, and were not used for feature selection.

## A Additional descriptions Development

| Variable | No target readmission (N = 182) | Target readmission (N = 130) | P value |
|---|---|---|---|
| LDH (U/L) | 381.00 (292.00, 519.75) | 401.50 (296.00, 545.25) | 0.460 |
| NEU (10^9/L) | 3.08 (1.84, 5.39); n=178 | 2.32 (1.50, 3.61); n=128 | 0.001 |
| LYM (10^9/L) | 0.76 (0.50, 1.15); n=178 | 0.65 (0.47, 0.95); n=128 | 0.026 |
| MON (10^9/L) | 0.32 (0.21, 0.48); n=178 | 0.27 (0.19, 0.44); n=128 | 0.071 |
| Potassium (mmol/L) | 3.90 (3.52, 4.20) | 3.80 (3.50, 4.10) | 0.222 |

| Variable | No target readmission (N = 182) | Target readmission (N = 130) | P value |
|---|---|---|---|
| Chloride (mmol/L) | 106.00 (104.00, 109.00) | 108.00 (104.00, 110.75) | 0.015 |
| PMR | 254.06 (169.15, 393.45); n=178 | 180.70 (131.34, 278.33); n=128 | <0.001 |
| PTAR | 0.42 (0.36, 0.49) | 0.52 (0.44, 0.62) | <0.001 |
| NAR | 1.01 (0.60, 1.84); n=178 | 0.83 (0.53, 1.39); n=128 | 0.040 |
| NLR | 3.54 (2.50, 6.67); n=178 | 3.56 (2.38, 6.18); n=128 | 0.551 |
| LMR | 2.44 (1.77, 3.50); n=178 | 2.38 (1.73, 3.44); n=128 | 0.696 |
| PLR | 102.17 (65.17, 156.44); n=178 | 75.50 (54.46, 115.70); n=128 | <0.001 |
| SIRI | 1.19 (0.57, 2.76); n=178 | 0.94 (0.49, 2.00); n=128 | 0.075 |
| PNI | 35.35 (31.52, 39.62); n=178 | 31.88 (28.84, 35.76); n=128 | <0.001 |
| APRI | 1.39 (0.78, 2.27) | 2.69 (1.56, 4.74) | <0.001 |

Values are medians (Q1, Q3), with available n when it differs from the group total. Outcome-group P values are exploratory, unadjusted, and were not used for feature selection.

## A Additional descriptions Internal validation

| Variable | No target readmission (N = 78) | Target readmission (N = 55) | P value |
|---|---|---|---|
| LDH (U/L) | 391.50 (282.00, 550.50) | 384.00 (267.00, 499.00) | 0.502 |
| NEU (10^9/L) | 2.53 (1.79, 4.72); n=77 | 3.03 (1.93, 4.58) | 0.663 |
| LYM (10^9/L) | 0.71 (0.46, 1.00); n=77 | 0.57 (0.46, 0.93) | 0.227 |
| MON (10^9/L) | 0.33 (0.22, 0.47); n=77 | 0.31 (0.17, 0.44) | 0.569 |
| Potassium (mmol/L) | 3.90 (3.52, 4.30) | 3.80 (3.60, 4.20) | 0.460 |
| Chloride (mmol/L) | 105.00 (103.00, 108.00) | 107.00 (105.00, 109.50) | 0.045 |
| PMR | 233.33 (168.42, 308.33); n=77 | 178.12 (114.95, 287.14) | 0.032 |
| PTAR | 0.42 (0.37, 0.48) | 0.55 (0.44, 0.69) | <0.001 |
| NAR | 0.90 (0.56, 1.39); n=77 | 1.08 (0.70, 1.85) | 0.129 |
| NLR | 3.71 (2.52, 7.11); n=77 | 4.70 (2.83, 7.74) | 0.295 |
| LMR | 2.33 (1.67, 3.33); n=77 | 2.25 (1.25, 3.40) | 0.600 |
| PLR | 107.04 (71.05, 153.13); n=77 | 78.43 (61.36, 119.92) | 0.021 |
| SIRI | 1.03 (0.65, 2.36); n=77 | 1.08 (0.67, 2.94) | 0.631 |
| PNI | 34.95 (30.90, 39.65); n=77 | 31.75 (28.30, 35.55) | 0.003 |

| Variable | No target readmission (N = 78) | Target readmission (N = 55) | P value |
|---|---|---|---|
| APRI | 1.38 (0.86, 2.66) | 2.21 (1.42, 3.60) | 0.006 |

Values are medians (Q1, Q3), with available n when it differs from the group total. Outcome-group P values are exploratory, unadjusted, and were not used for feature selection.

## A HBV DNA measurement status

| Status | Overall n | Analytical handling |
|---|---|---|
| Quantified result | 205 | Recorded continuous value and category |
| Left-censored with a recorded bound | 232 | Half of the recorded bound before log10 transformation |
| Not detected; numerical bound unavailable | 5 | Continuous value missing; original lowest-category assignment retained |
| No HBV DNA result | 3 | Continuous and categorical fields unavailable |

The 445 observations describe measurement and encoding states. Recorded left-censoring bounds were 20, 50, 100, and 1,000 IU/mL.

## A Viral-marker coding and limitations

| Field | Scale/definition | Measurement limitation |
|---|---|---|
| HBV DNA | log10 (IU/mL); categories <2,000, 2,000-<20,000 and ≥20,000 IU/mL | Continuous and categorical availability differ. Medians and rank-based tests do not remove uncertainty from censoring or substitution. |
| HBsAg | Stored log10 (IU/mL); available 445/445 | The historical 0.05-IU/mL convention and zero-to-0.025 coding are not verified assay-wide detection limits; no further substitution was applied. |

Pre-imputation categorical HBV DNA summaries use recorded categories; completed modeling categories use the completed $\log_{10}$ HBV DNA values.

## B Fixed-development selection status

| Predictor | LASSO λmin | LASSO λ1se | Original Boruta | After RoughFix | Core4 |
|---|---|---|---|---|---|
| Metabolic comorbidities | False | False | Rejected | False | False |
| EGVH | False | False | Rejected | False | False |
| Ascites | True | False | Rejected | False | False |
| SBP | True | False | Rejected | False | False |
| HRS | False | False | Rejected | False | False |
| Infection | True | False | Rejected | False | False |
| HBeAg status | True | False | Rejected | False | False |
| Sex | False | False | Rejected | False | False |
| Prior antiviral therapy | False | False | Rejected | False | False |

| Predictor | LASSO λmin | LASSO λ1se | Original Boruta | After RoughFix | Core4 |
|---|---|---|---|---|---|
| Smoking history | True | False | Rejected | False | False |
| Alcohol use history | True | False | Rejected | False | False |
| Scr | False | False | Rejected | False | False |
| TBIL | False | False | Confirmed | True | False |
| ALT | False | False | Rejected | False | False |
| AST | False | False | Rejected | False | False |
| GGT | False | False | Rejected | False | False |
| LDH | False | False | Rejected | False | False |
| ALP | False | False | Rejected | False | False |
| Albumin | False | False | Tentative | True | False |
| Hb | False | False | Rejected | False | False |
| NEU | False | False | Rejected | False | False |
| LYM | False | False | Rejected | False | False |
| MON | False | False | Rejected | False | False |
| PLT | True | True | Confirmed | True | True |
| WBC | False | False | Rejected | False | False |
| Potassium | False | False | Rejected | False | False |
| Sodium | False | False | Rejected | False | False |
| Chloride | False | False | Confirmed | True | False |
| INR | True | True | Confirmed | True | True |
| HBsAg | False | False | Rejected | False | False |
| HBV DNA (3 groups) | True | True | Rejected | False | False |
| HBV DNA (numeric) | False | False | Rejected | False | False |
| Age | True | False | Rejected | False | False |
| ALBI | True | True | Confirmed | True | True |
| PMR | False | False | Confirmed | True | False |
| PTAR | False | False | Confirmed | True | False |
| NAR | False | False | Rejected | False | False |
| NLR | False | False | Rejected | False | False |
| FIB-4 | True | True | Confirmed | True | True |
| LMR | False | False | Rejected | False | False |
| PLR | False | False | Rejected | False | False |

| Predictor | LASSO λmin | LASSO λ1se | Original Boruta | After RoughFix | Core4 |
|---|---|---|---|---|---|
| SIRI | False | False | Rejected | False | False |
| PNI | False | False | Rejected | False | False |
| APRI | False | False | Confirmed | True | False |
| Length of stay | False | False | Rejected | False | False |
| BMI | False | False | Rejected | False | False |

Both methods evaluated all 46 candidate source variables in the 312-patient development cohort, without filtering by univariable P values. Statuses refer to source variables rather than individual design-matrix terms. Initial Boruta results were nine confirmed, one tentative, and 36 rejected variables. Albumin was tentative and was retained after TentativeRoughFix. LASSO retained five variables at λıse, and Boruta retained ten; their intersection defined Core4.

## B Full nested-selection frequencies

| Predictor | Core4 reference set | Nested LASSO n (%) | Nested Boruta n (%) | Nested intersection n (%) |
|---|---|---|---|---|
| Metabolic comorbidities | No | 6 (6%) | 0 (0%) | 0 (0%) |
| EGVH | No | 9 (9%) | 0 (0%) | 0 (0%) |
| Ascites | No | 20 (20%) | 0 (0%) | 0 (0%) |
| SBP | No | 13 (13%) | 0 (0%) | 0 (0%) |
| HRS | No | 0 (0%) | 0 (0%) | 0 (0%) |
| Infection | No | 10 (10%) | 0 (0%) | 0 (0%) |
| HBeAg status | No | 10 (10%) | 0 (0%) | 0 (0%) |
| Sex | No | 11 (11%) | 0 (0%) | 0 (0%) |
| Prior antiviral therapy | No | 4 (4%) | 0 (0%) | 0 (0%) |
| Smoking history | No | 8 (8%) | 0 (0%) | 0 (0%) |
| Alcohol use history | No | 26 (26%) | 0 (0%) | 0 (0%) |
| Scr | No | 0 (0%) | 16 (16%) | 0 (0%) |
| TBIL | No | 0 (0%) | 99 (99%) | 0 (0%) |
| ALT | No | 0 (0%) | 6 (6%) | 0 (0%) |
| AST | No | 0 (0%) | 0 (0%) | 0 (0%) |
| GGT | No | 0 (0%) | 19 (19%) | 0 (0%) |
| LDH | No | 1 (1%) | 3 (3%) | 0 (0%) |
| ALP | No | 0 (0%) | 10 (10%) | 0 (0%) |
| Albumin | No | 0 (0%) | 50 (50%) | 0 (0%) |
| Hb | No | 0 (0%) | 0 (0%) | 0 (0%) |
| NEU | No | 0 (0%) | 2 (2%) | 0 (0%) |
| LYM | No | 0 (0%) | 10 (10%) | 0 (0%) |

| Predictor | Core4 reference set | Nested LASSO n (%) | Nested Boruta n (%) | Nested intersection n (%) |
|---|---|---|---|---|
| MON | No | 0 (0%) | 6 (6%) | 0 (0%) |
| PLT | Yes | 67 (67%) | 100 (100%) | 67 (67%) |
| WBC | No | 0 (0%) | 0 (0%) | 0 (0%) |
| Potassium | No | 0 (0%) | 1 (1%) | 0 (0%) |
| Sodium | No | 0 (0%) | 7 (7%) | 0 (0%) |
| Chloride | No | 1 (1%) | 56 (56%) | 1 (1%) |
| INR | Yes | 100 (100%) | 100 (100%) | 100 (100%) |
| HBsAg | No | 0 (0%) | 3 (3%) | 0 (0%) |
| HBV DNA (categorical) | No | 98 (98%) | 2 (2%) | 2 (2%) |
| log10 HBV DNA | No | 0 (0%) | 8 (8%) | 0 (0%) |
| Age | No | 0 (0%) | 1 (1%) | 0 (0%) |
| ALBI | Yes | 85 (85%) | 97 (97%) | 85 (85%) |
| PMR | No | 0 (0%) | 70 (70%) | 0 (0%) |
| PTAR | No | 5 (5%) | 100 (100%) | 5 (5%) |
| NAR | No | 0 (0%) | 0 (0%) | 0 (0%) |
| NLR | No | 0 (0%) | 0 (0%) | 0 (0%) |
| FIB-4 | Yes | 46 (46%) | 100 (100%) | 46 (46%) |
| LMR | No | 0 (0%) | 0 (0%) | 0 (0%) |
| PLR | No | 0 (0%) | 63 (63%) | 0 (0%) |
| SIRI | No | 0 (0%) | 2 (2%) | 0 (0%) |
| PNI | No | 0 (0%) | 42 (42%) | 0 (0%) |
| APRI | No | 1 (1%) | 100 (100%) | 1 (1%) |
| Length of stay | No | 1 (1%) | 0 (0%) | 0 (0%) |
| BMI | No | 0 (0%) | 0 (0%) | 0 (0%) |

All 100 outer training selections completed, and selection frequencies use all 100 regardless of subsequent model-fitting failures. Each outcome-stratified 70:30 partition comprised 312 outer training and 133 outer test patients. Selection used LASSO at $\lambda_{1se}$ and Boruta with TentativeRoughFix. Original Core4 membership served as a reference and did not restrict nested selection.

## C Post-selection Spearman correlations (development n = 312)

| Predictor | INR | PLT | ALBI | FIB-4 |
|---|---|---|---|---|
| INR | 1.000 | -0.351 | 0.518 | 0.434 |
| PLT | -0.351 | 1.000 | -0.143 | -0.744 |
| ALBI | 0.518 | -0.143 | 1.000 | 0.343 |
| FIB-4 | 0.434 | -0.744 | 0.343 | 1.000 |

## C Post-selection linear-redundancy diagnostics (development n = 312)

| Predictor | VIF | Tolerance |
|---|---|---|
| INR | 1.490 | 0.671 |
| Platelet count | 1.295 | 0.772 |
| ALBI | 1.407 | 0.711 |
| FIB-4 | 1.343 | 0.744 |

VIF = $1/(1 - R^2)$, obtained by regressing each Core4 predictor on the other three; tolerance = 1/VIF. These post-selection diagnostics were not selection criteria. They describe correlation and residual linear redundancy, rather than biological independence.

## C Joint Core4 retention and similarity

| Summary | Value | Definition |
|---|---|---|
| All four Core4 predictors retained | 35/100 (35%) | All four retained, allowing other predictors |
| Exact Core4 recovery | 33/100 (33%) | Selected intersection exactly Core4 |
| Jaccard similarity to original Core4 | 0.75 (0.50, 1.00) | Median (Q1, Q3); intersection size divided by union size |

Frequencies and quartiles describe behavior across repeated overlapping partitions; they are not independent-sample CIs or estimates of causal importance.

Abbreviations: EGVH, esophagogastric variceal hemorrhage; SBP, spontaneous bacterial peritonitis; HE, hepatic encephalopathy; HRS, hepatorenal syndrome; INR, international normalized ratio; PLT, platelet count; TBIL, total bilirubin; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ALP, alkaline phosphatase; GGT, gamma-glutamyl transferase; LDH, lactate dehydrogenase; PT, prothrombin time; PTA, prothrombin activity; WBC, white blood cell count; Hb, hemoglobin; Scr, serum creatinine; NEU, neutrophil count; LYM, lymphocyte count; MON, monocyte count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index; MELD-Na, Model for End-Stage Liver Disease-sodium; PMR, platelet-to-monocyte ratio; PTAR, study-defined 10 × INR/albumin; NAR, neutrophil-to-albumin ratio, calculated as 10 × NEU/albumin; NLR, neutrophil-to-lymphocyte ratio; LMR, lymphocyte-to-monocyte ratio; PLR, platelet-to-lymphocyte ratio; SIRI, systemic inflammation response index; PNI, prognostic nutritional index; APRI, aspartate aminotransferase-to-platelet ratio index; LASSO, least absolute shrinkage and selection operator; VIF, variance inflation factor. HBV, hepatitis B virus; DNA, deoxyribonucleic acid; HBeAg, hepatitis B e antigen; anti-HBe, antibody to hepatitis B e antigen; HBsAg, hepatitis B surface antigen; Q1/Q3, first/third quartiles.

## Table S4 Configuration of the seven candidate models

### A Fixed final model parameters and preprocessing

| Model | Fixed final parameters | Preprocessing | Development search |
|---|---|---|---|
| Logistic regression | link = logit; use_pred_offset = true | Not explicit in learner parameter set | No parameter search |
| LASSO | alpha = 1; lambda = 0.0232294425776311; standardize = true; use_pred_offset = true | Standardize = true (glmnet) | 10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC |
| Elastic net | alpha = 0.0842256563195503; lambda = 0.0893884745834839; standardize = true; use_pred_offset = true | Standardize = true (glmnet) | 10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC |
| Random forest | importance = none; min.node.size = 29; mtry.ratio = 0.2; num.threads = 1; num.trees = 1000; sample.fraction = 0.652165659546858 | Not explicit in learner parameter set | 10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC |
| XGBoost | alpha = 0; colsample_bytree = 1; eta = 0.0184324067769678; gamma = 0; lambda = 0.655478463773341; max_depth = 1; min_child_weight = 4.42216015579396; nrounds = 200; nthread = 1; subsample = 0.661144082310602; verbose = 0; verbosity = 0; use_pred_offset = true | Not explicit in learner parameter set | 10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC |
| RBF-SVM | cost = 0.274080705074942; gamma = 0.0573174496538081; kernel = radial; scale = true; type = C-classification | Scale = true | 10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC |
| LightGBM | num_leaves = 3; max_depth = 2; learning_rate = 0.014700166690139731; min_data_in_leaf = 50; feature_fraction = 0.75304694004165273; bagging_fraction = 0.70025569698636059; lambda_l2 = 19.999999999999996; nrounds = 169 | Identity on raw-scale Core4 values | 10-fold CV × 5 repeats (50 inner resamples); 30 MBO evaluations; optimize AUROC |

All fixed models used the 312-patient development cohort, with predictors ordered ALBI, FIB-4, INR, and PLT in the fitted learners. The positive class in the six original models was encoded as Yes. Tuning used development data alone. Glmnet standardization and SVM scaling used training-derived transformations; other preprocessing is reported only where specified in the learner settings. For RBF-SVM and Core4 logistic regression, probability prediction used the frozen wrapper with inputs matched by name; the SVM's stored probability-mapping parameters were retained.

Final parameter values describe the fixed development fits. Nested evaluation repeated selection, tuning, and fitting within each outer training partition, rather than imposing the fixed hyperparameters on every partition.

## B LightGBM search ranges

| Parameter | Search range | Fixed selected value |
|---|---|---|
| num_leaves | 2 to 15 (integer) | 3 |
| max_depth | 2 to 5 (integer) | 2 |
| learning_rate | 0.01 to 0.15 (log scale) | 0.014700166690139731 |
| min_data_in_leaf | 10 to 50 (integer) | 50 |
| feature_fraction | 0.7 to 1.0 | 0.75304694004165273 |
| bagging_fraction | 0.7 to 0.95 | 0.70025569698636059 |
| lambda_l2 | 0.5 to 20 (log scale) | 19.999999999999996 |
| nrounds | 50 to 400 (integer) | 169 |

## C LightGBM implementation and random settings

| Setting | Configuration |
|---|---|
| Implementation | LightGBM 4.7.0; binary objective; gradient-boosted decision trees; AUROC tuning metric |
| Inner evaluation | 10-fold cross-validation repeated 5 times (50 inner resamples); 30 MBO parameter evaluations |
| Execution | CPU; num_threads = 1; deterministic = true; force_col_wise = true; force_row_wise = false |
| Other fixed settings | bagging_freq = 1; feature_pre_filter = false; lambda_l1 = 0; min_gain_to_split = 0; extra_trees = false |
| Fixed tuning and final-fit seeds | Tuning: 700123; final fit: 1400123 |
| Native random seeds | seed: 800123; data: 900123; feature fraction: 1000123; bagging: 1100123; drop: 1200123; extra: 1300123 |

Statistical analyses primarily used R 4.5.1 and Python 3.12.14. CV, cross-validation; MBO, model-based optimization. AUROC, area under the receiver operating characteristic curve; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine; INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index. Core4 comprises INR, PLT, ALBI and FIB-4. SVM denotes RBF-SVM.

## Table S5 Supplementary model performance and clinical benchmarking

### A. Repeated nested internal performance

| Model | Valid n | AUROC | AP | Brier |
|---|---|---|---|---|
| Logistic regression | 98 | 0.776 (0.757, 0.797) | 0.693 (0.666, 0.723) | 0.192 (0.183, 0.201) |
| LASSO | 95 | 0.776 (0.759, 0.799) | 0.692 (0.668, 0.722) | 0.192 (0.184, 0.200) |
| Elastic net | 95 | 0.776 (0.756, 0.799) | 0.692 (0.662, 0.723) | 0.195 (0.188, 0.205) |
| Random forest | 98 | 0.764 (0.737, 0.781) | 0.645 (0.617, 0.682) | 0.196 (0.188, 0.209) |
| XGBoost | 98 | 0.766 (0.742, 0.780) | 0.661 (0.633, 0.692) | 0.195 (0.189, 0.205) |
| RBF-SVM | 98 | 0.777 (0.756, 0.798) | 0.688 (0.656, 0.719) | 0.193 (0.184, 0.201) |
| LightGBM | 98 | 0.762 (0.738, 0.779) | 0.653 (0.626, 0.688) | 0.199 (0.191, 0.209) |

Values are medians (Q1, Q3) across valid overlapping partitions and summarize resampling variability rather than CIs.

### A. Calibration in the same nested evaluations

| Model | Valid n | CITL | Calibration slope |
|---|---|---|---|
| Logistic regression | 98 | -0.009 (-0.114, 0.064) | 0.881 (0.718, 1.098) |
| LASSO | 95 | -0.007 (-0.108, 0.054) | 0.993 (0.802, 1.323) |
| Elastic net | 95 | -0.012 (-0.077, 0.046) | 1.196 (0.914, 1.869) |
| Random forest | 98 | -0.015 (-0.100, 0.077) | 0.861 (0.697, 1.023) |
| XGBoost | 98 | -0.025 (-0.087, 0.059) | 1.039 (0.850, 1.209) |
| RBF-SVM | 98 | -0.101 (-0.196, 0.005) | 0.932 (0.775, 1.107) |
| LightGBM | 98 | -0.020 (-0.076, 0.043) | 1.159 (0.879, 1.595) |

### A. Execution denominators

| Model | Planned | Eligible for primary comparison | Successful (primary set) | Failed (primary set) |
|---|---|---|---|---|
| Logistic regression | 100 | 98 | 98 | 0 |
| LASSO | 100 | 98 | 95 | 3 |
| Elastic net | 100 | 98 | 95 | 3 |
| Random forest | 100 | 98 | 98 | 0 |
| XGBoost | 100 | 98 | 98 | 0 |
| RBF-SVM | 100 | 98 | 98 | 0 |
| LightGBM | 100 | 98 | 98 | 0 |

Selection completed in all 100 planned partitions. Two partitions with unsupported categorical-predictor selections were excluded from the comparable set for all models. Three additional single-predictor partitions failed for LASSO and elastic net because their fitting interface required at least two columns.

Across all 100 planned partitions, successful/failed fits were 100/0 for logistic regression, 95/5 for LASSO, 95/5 for elastic net, 100/0 for random forest, 98/2 for XGBoost, 98/2 for RBF-SVM, and 100/0 for LightGBM. Completed fits excluded from the primary comparison are distinct from failed fits. Failures were not imputed, replaced, or retried.

### B. Fixed model performance and calibration: internal validation

| Metric | RBF-SVM | Core4 logistic | MELD-Na benchmark |
|---|---|---|---|
| AUROC | 0.751 | 0.762 | 0.691 |
| AP | 0.672 | 0.716 | 0.620 |
| Brier | 0.199 | 0.193 | 0.218 |
| CITL (95% CI) | 0.020 (-0.419 to 0.430) | -0.063 (-0.493 to 0.355) | -0.103 (-0.493 to 0.267) |
| Calibration slope (95% CI) | 0.822 (0.484 to 1.252) | 0.796 (0.480 to 1.268) | 0.863 (0.442 to 1.413) |
| O/E (95% CI) | 1.009 (0.827 to 1.199) | 0.973 (0.801 to 1.148) | 0.950 (0.770 to 1.133) |
| Mean predicted probability (95% CI) | 0.410 (0.369 to 0.452) | 0.425 (0.381 to 0.468) | 0.435 (0.405 to 0.466) |
| Observed event proportion (95% CI) | 0.414 (0.331 to 0.496) | 0.414 (0.331 to 0.496) | 0.414 (0.331 to 0.496) |

AUROC, AP, and Brier score are point estimates without marginal CIs in this panel. Calibration and prediction-summary intervals are reported where available.

### B. Fixed model performance and calibration: external evaluation

| Metric | RBF-SVM | Core4 logistic | MELD-Na benchmark |
|---|---|---|---|
| AUROC | 0.770 | 0.712 | 0.669 |
| AP | 0.853 | 0.785 | 0.732 |
| Brier | 0.199 | 0.225 | 0.214 |
| CITL (95% CI) | 0.725 (0.159 to 1.272) | 0.781 (0.085 to 1.458) | 0.055 (-0.730 to 0.682) |
| Calibration slope (95% CI) | 0.853 (0.415 to 1.491) | 0.223 (-0.038 to 0.774) | 0.412 (0.022 to 0.931) |
| O/E (95% CI) | 1.251 (1.058 to 1.447) | 1.231 (1.027 to 1.467) | 1.015 (0.811 to 1.178) |
| Mean predicted probability (95% CI) | 0.525 (0.432 to 0.606) | 0.533 (0.427 to 0.623) | 0.647 (0.575 to 0.717) |
| Observed event proportion (95% CI) | 0.657 (0.517 to 0.767) | 0.657 (0.517 to 0.767) | 0.657 (0.517 to 0.767) |

AUROC, AP, and Brier score are point estimates without marginal CIs in this panel. Calibration and prediction-summary intervals are reported where available.

### C. RBF-SVM versus MELD-Na: paired differences

| Cohort | AUROC difference (95% CI) | AP difference (95% CI) | Brier improvement (95% CI) |
|---|---|---|---|
| Internal validation | 0.060 (-0.021 to 0.141) | 0.052 (-0.050 to 0.159) | 0.019 (-0.011 to 0.048) |
| External evaluation | 0.100 (-0.029 to 0.233) | 0.122 (0.001 to 0.230) | 0.015 (-0.036 to 0.072) |

AUROC and AP differences are RBF-SVM minus MELD-Na; Brier improvement is MELD-Na minus RBF-SVM. Positive values favor RBF-SVM.

## C. Other paired comparisons: internal validation

| Comparison (left minus right) | AUROC difference (95% CI) | AP difference (95% CI) | Brier difference (95% CI) |
|---|---|---|---|
| Core4 logistic minus MELD-Na benchmark | 0.071 (-0.013 to 0.156) | 0.096 (0.001 to 0.195) | -0.025 (-0.054 to 0.005) |
| RBF-SVM minus Core4 logistic | -0.012 (-0.038 to 0.014) | -0.044 (-0.094 to 0.004) | 0.006 (-0.004 to 0.016) |

All differences are the left model minus the right model. A negative Brier difference favors the left model.

## C. Other paired comparisons: external evaluation

| Comparison (left minus right) | AUROC difference (95% CI) | AP difference (95% CI) | Brier difference (95% CI) |
|---|---|---|---|
| Core4 logistic minus MELD-Na benchmark | 0.042 (-0.082 to 0.158) | 0.054 (-0.043 to 0.140) | 0.011 (-0.041 to 0.065) |
| RBF-SVM minus Core4 logistic | 0.058 (-0.010 to 0.143) | 0.068 (-0.018 to 0.169) | -0.026 (-0.060 to -0.000) |

All differences are the left model minus the right model. A negative Brier difference favors the left model.

## D. Nested patient-level aggregate: coverage

| Patients | Events | Paired partitions | Total paired predictions |
|---|---|---|---|
| 445 | 185 | 98 | 13034 |

## D. Paired held-out predictions per patient

| Minimum | Q1 | Median | Mean | Q3 | Maximum |
|---|---|---|---|---|---|
| 15 | 26 | 29 | 29.2899 | 32 | 45 |

## D. Nested patient-level aggregate: absolute and paired performance

| Metric | RBF-SVM (95% CI) | MELD-Na benchmark (95% CI) | Paired difference (95% CI) |
|---|---|---|---|
| AUROC | 0.783 (0.739 to 0.824) | 0.722 (0.675 to 0.769) | 0.061 (0.025 to 0.097) |
| AP | 0.679 (0.606 to 0.760) | 0.618 (0.546 to 0.691) | 0.061 (0.012 to 0.116) |
| Brier | 0.189 (0.172 to 0.206) | 0.212 (0.196 to 0.228) | 0.024 (0.010 to 0.036) |

AUROC and AP differences are RBF-SVM minus MELD-Na; Brier improvement is MELD-Na minus RBF-SVM. Positive values favor RBF-SVM.

Each patient contributes the mean of paired predictions obtained only while held out. CIs use 5,000 paired patient-bootstrap samples and condition on the aggregated predictions without repeating model development. This aggregate summarizes the Chinese cohort rather than an independent cohort or a newly fitted single model.

Fixed internal validation comprised 133 patients with 55 events. External evaluation comprised 102 episodes from 50 patients, including 67 event episodes. AUROC, AP, and Brier score in panel B are point estimates; calibration and prediction summaries include their available intervals.

Fixed paired comparisons resample patients internally and patient clusters externally. Diagnostic calibration does not replace the evaluated predictions. Further details are in Notes M2 and M4.

Abbreviations: AUROC, area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large; CI, confidence interval; O/E, observed-to-expected ratio; Q1/Q3, first/third quartiles; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine; MELD-Na, Model for End-Stage Liver Disease-sodium. Core4 comprises international normalized ratio, platelet count, albumin-bilirubin score and fibrosis-4 index.

## E. External model-ranking stability under patient-cluster bootstrap resampling

### AUROC

| Model | Mean rank | First-place share (%) | Top-three share (%) |
|---|---|---|---|
| Logistic regression | 4.38 | 1.13 | 28.19 |
| LASSO | 5.41 | 0.11 | 13.72 |
| Elastic net | 3.59 | 6.40 | 48.75 |
| Random forest | 4.11 | 0.31 | 49.78 |
| XGBoost | 3.31 | 4.43 | 62.11 |
| RBF-SVM | 1.24 | 87.62 | 95.61 |
| LightGBM | 5.96 | 0.00 | 1.84 |

### AP

| Model | Mean rank | First-place share (%) | Top-three share (%) |
|---|---|---|---|
| Logistic regression | 4.86 | 1.00 | 14.76 |
| LASSO | 6.16 | 0.02 | 5.40 |
| Elastic net | 4.34 | 6.36 | 25.00 |
| Random forest | 3.36 | 1.78 | 73.60 |
| XGBoost | 2.24 | 28.74 | 86.92 |
| RBF-SVM | 1.73 | 62.10 | 91.80 |
| LightGBM | 5.31 | 0.00 | 2.52 |

### Brier score

| Model | Mean rank | First-place share (%) | Top-three share (%) |
|---|---|---|---|
| Logistic regression | 5.16 | 0.80 | 12.36 |
| LASSO | 5.42 | 0.14 | 14.06 |
| Elastic net | 4.11 | 4.60 | 32.60 |
| Random forest | 3.55 | 0.92 | 61.00 |
| XGBoost | 3.35 | 2.80 | 65.62 |
| RBF-SVM | 1.20 | 89.54 | 97.36 |
| LightGBM | 5.21 | 1.20 | 17.00 |

Rank summaries use 5,000 common patient-cluster bootstrap samples of fixed external predictions for 102 episodes from 50 patients (67 event episodes). Each sample draws 50 patients with replacement and retains all their eligible episodes with the corresponding multiplicities; estimates remain episode-weighted. Models are ranked separately by higher AUROC, higher AP, and lower Brier score, with 1 indicating the best rank. Tied ranks are equally apportioned using a tolerance of $10^{-12}$. Mean rank is averaged across samples; first-place and top-three shares sum the corresponding apportioned frequencies. All 5,000 samples were valid for each metric. Shares describe resampling rankings and are not probabilities that a model is truly optimal.

Abbreviations: AUROC, area under the receiver operating characteristic curve; AP, average precision; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.

## F. Paired external performance differences between RBF-SVM and the other candidate models

### ΔAUROC (RBF-SVM minus comparator)

| Comparator | Point difference (95% bootstrap CI) | RBF-SVM-leading frequency (%) |
|---|---|---|
| Logistic regression | +0.0580 (-0.0104 to 0.1435) | 94.68 |
| LASSO | +0.0648 (-0.0068 to 0.1551) | 95.94 |
| Elastic net | +0.0525 (-0.0167 to 0.1381) | 91.94 |
| Random forest | +0.0503 (0.0042 to 0.1173) | 98.66 |
| XGBoost | +0.0420 (-0.0083 to 0.1112) | 94.32 |
| LightGBM | +0.0753 (0.0221 to 0.1458) | 99.80 |

### ΔAP (RBF-SVM minus comparator)

| Comparator | Point difference (95% bootstrap CI) | RBF-SVM-leading frequency (%) |
|---|---|---|
| Logistic regression | +0.0678 (-0.0182 to 0.1691) | 91.94 |
| LASSO | +0.0774 (-0.0149 to 0.1793) | 93.80 |
| Elastic net | +0.0656 (-0.0231 to 0.1680) | 90.08 |
| Random forest | +0.0193 (-0.0176 to 0.0948) | 85.08 |
| XGBoost | +0.0044 (-0.0326 to 0.0884) | 67.32 |
| LightGBM | +0.0544 (0.0056 to 0.1429) | 99.00 |

### Brier improvement (comparator minus RBF-SVM)

| Comparator | Point difference (95% bootstrap CI) | RBF-SVM-leading frequency (%) |
|---|---|---|
| Logistic regression | +0.0256 (0.0000243 to 0.0600) | 97.52 |
| LASSO | +0.0262 (-0.0015 to 0.0608) | 96.66 |
| Elastic net | +0.0224 (-0.0057 to 0.0555) | 94.20 |
| Random forest | +0.0161 (0.0003 to 0.0336) | 97.64 |
| XGBoost | +0.0156 (-0.0021 to 0.0349) | 95.94 |
| LightGBM | +0.0264 (0.0019 to 0.0483) | 98.16 |

Point differences use the complete external cohort. Paired 95% percentile CIs are the 2.5th and 97.5th percentiles of within-sample differences from the same 5,000 patient-cluster samples as panel E. Differences are RBF-SVM minus comparator for AUROC and AP and comparator minus RBF-SVM for Brier score; positive values favor RBF-SVM. The RBF-SVM-leading frequency is the proportion of valid samples with a difference $>10^{-12}$, retaining ties in the denominator; it is not a P value. All samples were valid. CIs are comparison-specific and unadjusted for multiplicity; additional digits are retained for near-zero bounds. Models were not refitted. These exploratory summaries do not provide independent validation or adjust for primary-model selection.

Abbreviations: CI, confidence interval; AUROC, area under the receiver operating characteristic curve; AP, average precision; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.

## Table S6 Predictor representation and Core4 ablation analyses

### A. Internal validation: Logistic regression

| Representation | AUROC | AP | Brier | CITL | Slope |
|---|---|---|---|---|---|
| Core4 reference set | 0.762 | 0.716 | 0.193 | -0.063 | 0.796 |
| Raw-only intersection | 0.755 | 0.715 | 0.197 | -0.118 | 0.762 |
| Composite representation | 0.744 | 0.676 | 0.202 | 0.064 | 0.926 |

### A. Internal validation: RBF-SVM

| Representation | AUROC | AP | Brier | CITL | Slope |
|---|---|---|---|---|---|
| Core4 reference set | 0.751 | 0.672 | 0.199 | 0.020 | 0.822 |
| Raw-only intersection | 0.763 | 0.724 | 0.191 | -0.064 | 0.873 |
| Composite representation | 0.739 | 0.667 | 0.204 | 0.044 | 0.970 |

### A. External evaluation: Logistic regression

| Representation | AUROC | AP | Brier | CITL | Slope |
|---|---|---|---|---|---|
| Core4 reference set | 0.712 | 0.785 | 0.225 | 0.781 | 0.223 |
| Raw-only intersection | 0.665 | 0.744 | 0.249 | 0.825 | 0.124 |
| Composite representation | 0.719 | 0.797 | 0.214 | 0.601 | 0.420 |

### A. External evaluation: RBF-SVM

| Representation | AUROC | AP | Brier | CITL | Slope |
|---|---|---|---|---|---|
| Core4 reference set | 0.770 | 0.853 | 0.199 | 0.725 | 0.853 |
| Raw-only intersection | 0.675 | 0.765 | 0.243 | 0.684 | 0.240 |
| Composite representation | 0.733 | 0.834 | 0.209 | 0.582 | 0.755 |

### B. Core4 ablations: internal validation / Logistic regression

| Ablation | Delta AUROC (95% CI) | Delta AP (95% CI) | Brier improvement (95% CI) |
|---|---|---|---|
| E1: remove INR | 0.030 (-0.007 to 0.067) | 0.063 (0.004 to 0.117) | 0.015 (0.001 to 0.029) |
| E2: remove PLT | -0.006 (-0.043 to 0.029) | -0.002 (-0.037 to 0.034) | 0.001 (-0.012 to 0.013) |
| E3: remove ALBI | 0.011 (-0.015 to 0.037) | -0.001 (-0.035 to 0.034) | 0.005 (-0.004 to 0.014) |
| E4: remove FIB-4 | -0.011 (-0.025 to 0.002) | -0.016 (-0.046 to 0.009) | -0.002 (-0.008 to 0.003) |
| E5: INR + ALBI only | -0.024 (-0.078 to 0.025) | -0.029 (-0.080 to 0.019) | -0.008 (-0.027 to 0.010) |
| E6: PLT + FIB-4 only | 0.095 (0.037 to 0.158) | 0.149 (0.058 to 0.224) | 0.034 (0.013 to 0.054) |

## B. Core4 ablations: internal validation / RBF-SVM

| Ablation | Delta AUROC (95% CI) | Delta AP (95% CI) | Brier improvement (95% CI) |
|---|---|---|---|
| E1: remove INR | 0.037 (-0.004 to 0.078) | 0.056 (-0.011 to 0.118) | 0.016 (-0.002 to 0.033) |
| E2: remove PLT | -0.003 (-0.025 to 0.017) | -0.023 (-0.075 to 0.028) | -0.002 (-0.010 to 0.007) |
| E3: remove ALBI | 0.010 (-0.022 to 0.043) | -0.026 (-0.079 to 0.028) | 0.004 (-0.008 to 0.017) |
| E4: remove FIB-4 | -0.024 (-0.053 to 0.005) | -0.043 (-0.110 to 0.020) | -0.010 (-0.024 to 0.003) |
| E5: INR + ALBI only | -0.036 (-0.082 to 0.008) | -0.072 (-0.139 to -0.005) | -0.012 (-0.031 to 0.006) |
| E6: PLT + FIB-4 only | 0.090 (0.026 to 0.155) | 0.114 (0.022 to 0.196) | 0.032 (0.008 to 0.056) |

## B. Core4 ablations: external evaluation / Logistic regression

| Ablation | Delta AUROC (95% CI) | Delta AP (95% CI) | Brier improvement (95% CI) |
|---|---|---|---|
| E1: remove INR | -0.049 (-0.133 to 0.030) | -0.084 (-0.172 to 0.008) | 0.002 (-0.041 to 0.040) |
| E2: remove PLT | 0.005 (-0.042 to 0.055) | 0.013 (-0.026 to 0.058) | -0.007 (-0.028 to 0.014) |
| E3: remove ALBI | 0.032 (0.006 to 0.065) | 0.033 (0.004 to 0.070) | 0.012 (-0.001 to 0.028) |
| E4: remove FIB-4 | 0.008 (-0.010 to 0.029) | 0.010 (-0.011 to 0.035) | 0.009 (-0.001 to 0.017) |
| E5: INR + ALBI only | 0.056 (-0.021 to 0.138) | 0.058 (-0.012 to 0.126) | 0.018 (-0.018 to 0.053) |
| E6: PLT + FIB-4 only | -0.001 (-0.105 to 0.100) | -0.053 (-0.145 to 0.055) | 0.013 (-0.034 to 0.055) |

## B. Core4 ablations: external evaluation / RBF-SVM

| Ablation | Delta AUROC (95% CI) | Delta AP (95% CI) | Brier improvement (95% CI) |
|---|---|---|---|
| E1: remove INR | -0.004 (-0.059 to 0.055) | -0.022 (-0.054 to 0.032) | -0.003 (-0.028 to 0.022) |
| E2: remove PLT | 0.025 (-0.035 to 0.099) | 0.016 (-0.045 to 0.123) | 0.003 (-0.019 to 0.031) |
| E3: remove ALBI | 0.072 (-0.007 to 0.166) | 0.083 (-0.018 to 0.189) | 0.016 (-0.017 to 0.061) |
| E4: remove FIB-4 | 0.016 (-0.028 to 0.061) | 0.006 (-0.034 to 0.055) | 0.010 (-0.009 to 0.027) |
| E5: INR + ALBI only | 0.107 (0.009 to 0.221) | 0.105 (-0.010 to 0.218) | 0.038 (-0.001 to 0.079) |
| E6: PLT + FIB-4 only | 0.034 (-0.039 to 0.125) | -0.003 (-0.041 to 0.082) | 0.009 (-0.022 to 0.044) |

Panel A reports point estimates for fixed internal validation (133 patients) and external evaluation (102 episodes from 50 patients). Selected intersections were INR/PLT/ALBI/FIB-4 for Core4, INR/PLT for raw-variable candidates, and PTAR/FIB-4 for composite candidates. Candidate-pool definitions are in Note M3.

Panel B reports all six ablations. ΔAUROC and ΔAP are full model minus ablation; Brier improvement is ablation minus full model. Positive values favor the full model. Paired 95% percentile CIs use 5,000 patient-bootstrap samples internally and patient-cluster samples externally. Intervals crossing zero do not establish equivalence or absence of contribution. Details are in Note M3.

Abbreviations: AUROC, area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large; CI, confidence interval; RBF-SVM, radial-basis-function support vector machine; PTAR, study-defined 10 × INR/albumin (albumin in g/L); INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index.

## Table S7 Discrimination by readmission timing

### A. Event-time groups and shared non-event controls

| Cohort / period | Cases | Common controls | Analyzed records | Patients / units | Unclassified positive times |
|---|---|---|---|---|---|
| Internal / Early | 22 | 78 | 100 | 100 | 0 |
| Internal / Late | 33 | 78 | 111 | 111 | 0 |
| External / Early | 48 | 35 | 83 | 48 | 2 |
| External / Late | 17 | 35 | 52 | 37 | 2 |

Internal units are patients, and external units are hospitalization episodes. The same 1-year non-event controls are shared by the early and later comparisons; overlapping comparison sizes cannot be added to reconstruct the primary cohort.

Two external event episodes with unclassifiable timing remain in the annual analysis but are excluded from both timing groups. The repeated count in the external rows refers to the same two episodes.

### B. Fixed internal model scores: time-stratified discrimination

| Model | Early AUROC (95% CI) | Late AUROC (95% CI) | Late minus early (95% CI) |
|---|---|---|---|
| Logistic regression | 0.846 (0.745 to 0.930) | 0.707 (0.591 to 0.814) | -0.139 (-0.270 to -0.012) |
| LASSO | 0.848 (0.749 to 0.932) | 0.711 (0.596 to 0.817) | -0.137 (-0.267 to -0.011) |
| Elastic net | 0.843 (0.744 to 0.929) | 0.709 (0.594 to 0.814) | -0.134 (-0.261 to -0.009) |
| Random forest | 0.837 (0.736 to 0.924) | 0.711 (0.601 to 0.815) | -0.126 (-0.253 to -0.004) |
| XGBoost | 0.838 (0.736 to 0.926) | 0.686 (0.572 to 0.793) | -0.152 (-0.285 to -0.026) |
| RBF-SVM | 0.818 (0.712 to 0.911) | 0.706 (0.593 to 0.811) | -0.112 (-0.235 to 0.012) |

### C. Fixed external model scores: time-stratified discrimination

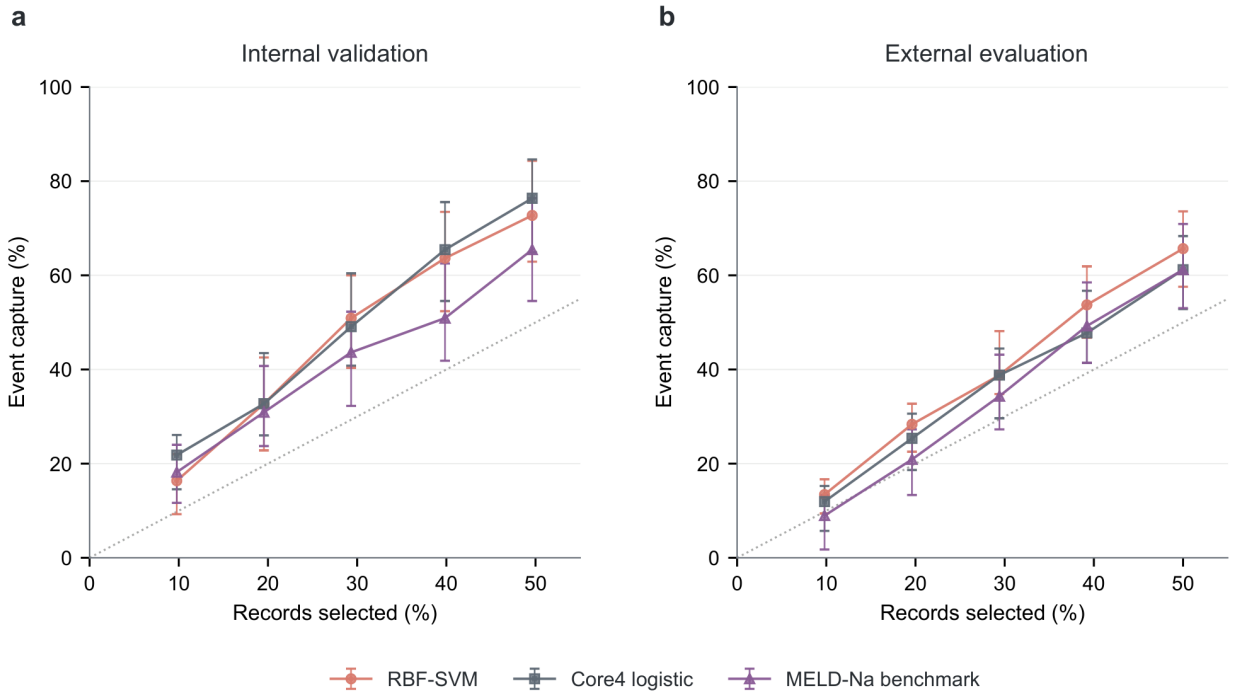
| Model | Early AUROC (95% CI) | Late AUROC (95% CI) | Late minus early (95% CI) |
|---|---|---|---|
| Logistic regression | 0.725 (0.583 to 0.835) | 0.645 (0.438 to 0.803) | -0.080 (-0.243 to 0.048) |
| LASSO | 0.718 (0.577 to 0.829) | 0.640 (0.436 to 0.798) | -0.078 (-0.241 to 0.051) |
| Elastic net | 0.732 (0.592 to 0.842) | 0.649 (0.444 to 0.807) | -0.083 (-0.255 to 0.047) |
| Random forest | 0.728 (0.574 to 0.847) | 0.684 (0.447 to 0.853) | -0.044 (-0.232 to 0.112) |
| XGBoost | 0.732 (0.569 to 0.853) | 0.691 (0.456 to 0.857) | -0.042 (-0.227 to 0.109) |
| RBF-SVM | 0.786 (0.655 to 0.887) | 0.723 (0.494 to 0.878) | -0.064 (-0.266 to 0.077) |

Early events occurred at 0–90 days, and later events at >90–365 days. Each event group is compared with the same 1-year non-event controls; the other event group is excluded. These analyses evaluate six fixed annual model scores, rather than newly fitted early/later models, cumulative 90-day ROC curves, or time-specific calibrated risks.

CIs and later-minus-earlier contrasts use 5,000 paired resamples of patients internally and patient clusters externally. An interval crossing zero does not establish equivalence. Details are in Note M3.

Abbreviations: AUROC, area under the receiver operating characteristic (ROC) curve; CI, confidence interval; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting.

**Fig. S1 Observed-event capture at matched selection capacity**




(a) Fixed internal validation (133 patients) and (b) external evaluation (102 hospitalization episodes from 50 patients). RBF-SVM, Core4 logistic regression, and the MELD-Na benchmark are compared at nominal selection proportions of 10%, 20%, 30%, 40%, and 50%. Each selects the K highest-scoring observations, where K = floor(qN), q is nominal capacity, and N is the cohort record count. The horizontal axis shows actual coverage (K/N); the vertical axis shows the proportion of all observed annual event records captured. Points and bars are estimates and pointwise 95% percentile CIs from 5,000 resamples of patients internally and patient clusters externally. Top K selection is repeated within each resample; ties use the fixed outcome-blind identifier hash. Lines join the five scenarios; the dotted diagonal shows expected capture under random ranking. Core4 comprises INR, PLT, ALBI, and FIB-4. MELD-Na is a development-fitted single-predictor logistic benchmark. These hypothetical allocation scenarios differ from the development-derived risk strata. Intervals condition on fixed predictions, and capture does not estimate prevented events or an intervention effect

RBF-SVM, radial-basis-function support vector machine; MELD-Na, Model for End-Stage Liver Disease-sodium; CI, confidence interval; INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index

## Fig. S2 Full-range decision curves and paired net-benefit differences

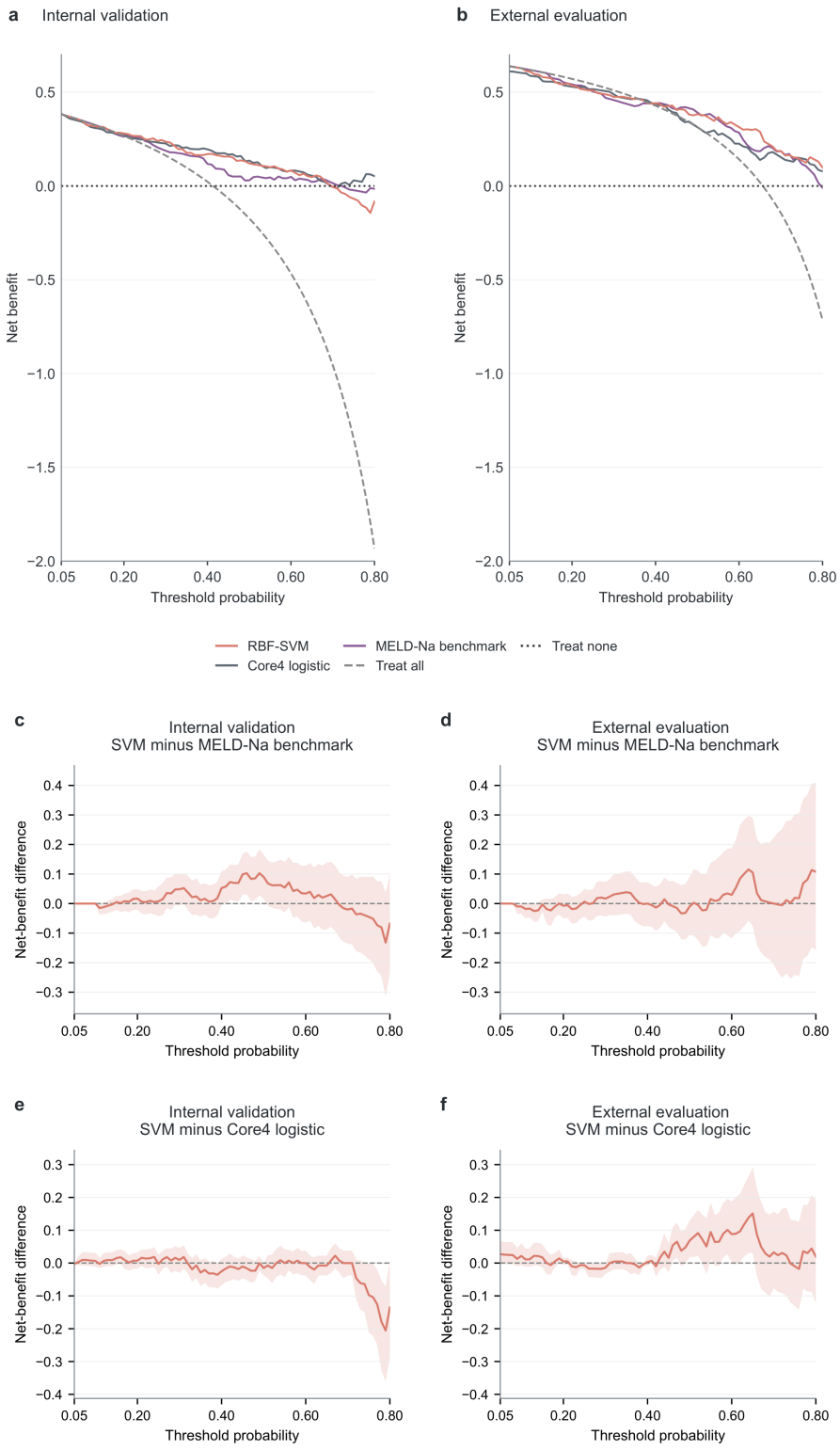



**Fig. S2** (a, b) Decision curves in fixed internal validation (133 patients) and external evaluation (102 episodes from 50 patients). Solid curves show net benefit for RBF-SVM, Core4 logistic regression, and the MELD-Na benchmark; dashed gray and dotted zero lines denote treat-all and treat-none. These panels use identical axes and show point estimates. (c, d) RBF-SVM minus MELD-Na net-benefit differences internally and externally. (e, f) Corresponding RBF-SVM minus Core4 logistic-regression differences. All panels use thresholds from 0.05 to 0.80 at increments of 0.01. Shading in c–f shows pointwise 95% percentile CIs of paired differences from 5,000 patient resamples internally and patient-cluster resamples externally. Dashed

horizontal lines denote zero; positive differences favor RBF-SVM at that threshold. Intervals are pointwise rather than simultaneous across thresholds. Core4 comprises INR, PLT, ALBI, and FIB-4; the MELD-Na benchmark is a development-fitted single-predictor logistic model. Net benefit weights false-positive classifications by threshold odds. Comparisons condition on fixed predictions, without evaluation-cohort recalibration or threshold optimization. Treat-all and treat-none refer to a hypothetical risk-triggered action; the displayed range does not establish validated intervention thresholds or demonstrated clinical benefit

DCA, decision-curve analysis; RBF-SVM, radial-basis-function support vector machine (SVM in the panels); MELD-Na, Model for End-Stage Liver Disease-sodium; CI, confidence interval; INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index

## Table S8 Baseline characteristics of the Chinese cohort

| Characteristic | Overall cohort (n = 445) |
|---|---|
| Demographic and background characteristics | |
| Age (years) | 52.0 (45.0, 61.0) |
| Male sex | 313 (70.3) |
| BMI (kg/m²) | 22.49 (20.55, 24.44) |
| Obesity, n (%) | 35 (7.9) |
| Hypertension, n (%) | 51 (11.5) |
| Diabetes mellitus, n (%) | 79 (17.8) |
| Metabolic comorbidities, n (%) | 131 (29.4) |
| Smoking history | 185 (41.6) |
| Alcohol use history | 183 (41.1) |
| HBV serological, virological and treatment-related characteristics | |
| HBeAg/anti-HBe status | |
| HBeAg−/anti-HBe+ | 313 (70.3) |
| HBeAg+/anti-HBe− | 77 (17.3) |
| HBeAg−/anti-HBe− | 36 (8.1) |
| HBeAg+/anti-HBe+ | 19 (4.3) |
| HBV DNA study category (IU/mL) | Available n = 442 |
| <2,000 | 260 (58.8) |
| 2,000–<20,000 | 47 (10.6) |
| ≥20,000 | 135 (30.5) |
| HBV DNA, log10(IU/mL) | 2.70 (2.70, 4.97); n = 437 |
| HBsAg, log10(IU/mL) | 2.81 (1.94, 3.16) |
| Prior antiviral therapy | 246 (55.3) |
| Recorded clinical manifestations | |
| Ascites | 194 (43.6) |
| EGVH | 246 (55.3) |
| SBP | 108 (24.3) |
| HE | 27 (6.1) |
| HRS | 5 (1.1) |

| Characteristic | Overall cohort (n = 445) |
|---|---|
| Infection | 181 (40.7) |
| Laboratory measurements | |
| Albumin (g/L) | 29.00 (26.00, 34.00) |
| Total bilirubin (μmol/L) | 24.70 (15.70, 41.00) |
| ALT (U/L) | 39.00 (27.00, 57.00) |
| AST (U/L) | 44.00 (30.00, 65.00) |
| ALP (U/L) | 92.00 (67.00, 128.00) |
| GGT (U/L) | 38.00 (24.00, 74.00) |
| INR | 1.37 (1.22, 1.52) |
| PT (s) | 16.40 (15.10, 18.30) |
| PTA (%) | 61.00 (52.00, 71.00) |
| Platelet count ($10^9$/L) | 63.00 (44.00, 95.00) |
| WBC ($10^9$/L) | 3.97 (2.70, 5.95) |
| Hemoglobin (g/L) | 92.00 (73.00, 111.00) |
| Serum creatinine (μmol/L) | 67.00 (56.00, 79.00) |
| Sodium (mmol/L) | 139.00 (137.00, 141.00) |
| Composite scores | |
| MELD-Na | 12.47 (9.99, 15.54) |
| ALBI | -1.62 (-2.02, -1.22) |
| FIB-4 | 5.76 (3.38, 10.53) |
| Index hospitalization | |
| Length of index hospital stay (days) | 10.0 (7.0, 15.0) |
| Observed 1-year outcome | |
| Target readmission | 185 (41.6) |

Values are medians (interquartile ranges) or n (%), calculated from available nonmissing observations. Available n is shown when it differs from 445. Clinical manifestations can coexist. Metabolic comorbidity denotes obesity, hypertension, or diabetes mellitus; obesity was defined as BMI ≥28 kg/m². Viral-marker coding, measurement limitations, and derived-index definitions are detailed in Table S3A and Note M1.

Abbreviations: ALBI, albumin-bilirubin score; ALP, alkaline phosphatase; ALT, alanine aminotransferase; anti-HBe, antibody to hepatitis B e antigen; AST, aspartate aminotransferase; BMI, body mass index; DNA, deoxyribonucleic acid; EGVH, esophagogastric variceal hemorrhage; FIB-4, fibrosis-4 index; GGT, gamma-glutamyl transferase; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HE, hepatic encephalopathy; HRS, hepatorenal syndrome; INR, international normalized ratio; MELD-Na, Model for End-Stage Liver Disease incorporating serum sodium; PT, prothrombin time; PTA, prothrombin activity; SBP, spontaneous bacterial peritonitis; WBC, white blood cell count.

## Table S9 Coverage and observed-event capture across RBF-SVM risk strata

| Cohort | Risk-stratum strategy | Selected records, n/N (%) | Captured events, n/total events (%) | Missed events, n | Specificity, % (95% CI) |
|---|---|---|---|---|---|
| Internal validation | High only | 39/133 (29.3%) | 28/55 (50.9%) | 27 | 85.9 (77.6–93.0) |
| Internal validation | Intermediate plus High | 89/133 (66.9%) | 48/55 (87.3%) | 7 | 47.4 (36.2–58.4) |
| External evaluation | High only | 52/102 (51.0%) | 45/67 (67.2%) | 22 | 80.0 (65.8–92.5) |
| External evaluation | Intermediate plus High | 81/102 (79.4%) | 59/67 (88.1%) | 8 | 37.1 (20.6–54.8) |

Risk strata use the lower and upper tertiles of development-cohort RBF-SVM predictions, applied unchanged to both evaluation cohorts; exact cutoffs and boundary conventions are in Note M4. Coverage is the proportion of observations selected; event capture is the proportion of all observed target events among selected observations. Specificity is the proportion of non-event observations outside the selected group. External results refer to 102 episodes from 50 patients. Captured events do not estimate prevented readmissions, and these quantile cutoffs are not validated intervention thresholds.

Abbreviations: CI, confidence interval; RBF-SVM, radial-basis-function support vector machine.

## Note M6 Extended methods

### Study design and population

This single-center retrospective cohort study included patients with HBV-related decompensated cirrhosis hospitalized at the First Affiliated Hospital of Chongqing Medical University between January 2012 and December 2023. Discharge was the prediction time point, and the prediction horizon was the subsequent year.

HBV-related cirrhosis was identified according to the Guidelines for the prevention and treatment of chronic hepatitis B (version 2022) [9], using HBV serology or a documented history of chronic HBV infection together with clinical or imaging evidence of cirrhosis. Index hospitalization records were manually reviewed for eligibility, which required qualifying decompensation. Patients with other major causes of liver disease were excluded. Other exclusion criteria included planned index hospitalization, alcohol-related cirrhosis, autoimmune liver disease, tumors, liver transplantation, incomplete follow-up, and extensive missing clinical data (Fig. 1). Acute-on-chronic liver failure (ACLF) was assessed retrospectively against the Guidelines for the diagnosis and treatment of acute-on-chronic liver failure (2025 version) [10]; patients meeting these criteria were excluded.

The study sample comprised all Chinese patients and external hospitalization episodes meeting the study eligibility criteria. The final Chinese cohort comprised 445 patients and was divided in an outcome-stratified 70:30 ratio into development (n = 312) and fixed internal validation (n = 133) cohorts. The study was conducted in accordance with the Declaration of Helsinki. The Medical Research Ethics Review Committee of the First Affiliated Hospital of Chongqing Medical University approved the study and the application for a waiver of informed consent (approval no. 2025-925-01; November 21, 2025).

### Outcome and follow-up

The primary outcome was unplanned readmission within 1 year after discharge for variceal bleeding, ascites, hepatic encephalopathy, hepatorenal syndrome (HRS), or spontaneous bacterial peritonitis. Readmissions for planned treatment, investigations, or other elective arrangements were not outcome events.

Readmissions were ascertained through inpatient records, outpatient records, and telephone follow-up through 1 year after discharge, with manual review of the records. Patients with incomplete follow-up and those who died during this period before a target readmission were excluded from the final binary analysis.

All authors participated in reviewing eligibility and target-readmission classification in the Chinese cohort. Disagreements were resolved through discussion and consensus. Reviewers in both cohorts did not see model predictions during outcome assessment.

### Predictor measurement and missing data

We extracted 46 predefined candidate predictors from electronic medical records, covering demographic and lifestyle characteristics, HBV serological and virological measures, cirrhosis-related clinical characteristics, routine laboratory tests, and derived indices (Table S3). Laboratory values used for modeling and clinical benchmarking were the available measurements closest to admission, primarily from admission or the early inpatient period. Variable definitions, measurement scales, analytical roles, and pre-imputation availability are detailed in Table S3 and Note M1.

Predictor data in the Chinese cohort were extracted from medical records and checked by infectious disease clinicians involved in the study. In both cohorts, personnel extracting and checking predictor data did not see subsequent readmission outcomes.

After the development–validation split, the small amount of missing data in the Chinese cohort was handled by multiple imputation by chained equations. Imputation parameters were estimated from the development cohort alone; internal validation data did not contribute to parameter estimation. The outcome was neither imputed nor used as an imputation predictor. Five completed datasets were generated, and one dataset fixed

in advance was used for subsequent model development and evaluation. Derived indices were recalculated after processing their component variables, using the formulas and rounding rules in Table S3A. Estimates were not pooled across completed datasets.

## External evaluation cohort

The external cohort was drawn from the Medical Information Mart for Intensive Care IV (MIMIC-IV), version 3.1 [11,12]. Two experienced infectious disease clinicians independently reviewed hospitalization eligibility and target-readmission classification for the external cohort. They applied the clinical inclusion and exclusion criteria and guideline-based outcome definitions used for the Chinese cohort, integrating available diagnoses, investigation reports, and medical orders recorded in MIMIC-IV. Disagreements were resolved through discussion and consensus. For the 1-year postdischarge assessment, subsequent encounter records and available mortality information in MIMIC-IV were reviewed. Eligible hospitalization episodes with no target readmission documented in MIMIC-IV within 1 year after discharge were classified as non-events. Readmission ascertainment was limited to events documented in the database. The analysis unit was an eligible hospitalization episode. A patient could contribute multiple episodes, each with a separately defined 1-year postdischarge outcome window. Model inputs were observed measurements from the same continuous emergency–inpatient encounter, selected as close to admission as available data allowed; no statistical imputation was performed. Note M1 describes the measurement hierarchy and its exceptions. Analyses used episode-level predictions, with patient clustering accounted for in uncertainty estimation.

## Feature selection

Categorical variables were encoded using predefined clinical categories, and continuous variables retained their numerical form, with the viral-marker transformations described in Table S3A. Feature selection used only development data. Least absolute shrinkage and selection operator (LASSO) regression and Boruta were applied independently to all 46 candidates; their intersection defined the primary predictor set. LASSO used tenfold cross-validation and the one-standard-error penalty ($\lambda_{1se}$), while TentativeRoughFix resolved tentative Boruta selections. The resulting four-measure set, termed Core4, comprised the international normalized ratio (INR), platelet count (PLT), albumin-bilirubin (ALBI) score, and fibrosis-4 (FIB-4) index. Spearman correlations and variance inflation factors (VIFs) were examined after selection, rather than used as selection criteria (Table S3C).

Selection stability was assessed by repeating missing-data handling, LASSO, and Boruta within each outer training set during repeated nested evaluation. We recorded selection frequencies and the Jaccard similarity between each selected intersection and the original Core4 set. Exploratory analyses compared alternative predictor representations and six Core4 ablations (Table S6).

## Candidate models and evaluation strategy

Seven models were fitted using the common selected predictor set: unpenalized logistic regression, LASSO, elastic net, random forest, XGBoost, LightGBM, and a radial-basis-function support vector machine (RBF-SVM). Hyperparameter optimization was confined to development data and used tenfold cross-validation repeated five times, with 30 model-based optimization evaluations and area under the receiver operating characteristic curve (AUROC) as the tuning criterion. Unpenalized logistic regression did not undergo a parameter search. Final settings, preprocessing, and available search configurations are reported in Table S4.

Performance was assessed using AUROC, average precision (AP), the Brier score, calibration-in-the-large (CITL), and calibration slope. To assess the full modeling procedure under sample perturbation, we conducted 100 outcome-stratified repeated nested partitions of the Chinese cohort, each with 312 outer training and 133 outer test patients. Missing-data handling, feature selection, hyperparameter optimization, and fitting were repeated within the outer training data. Outer test observations did not inform imputation parameters, feature selection, or hyperparameter choice.

All 100 selection procedures completed. The primary seven-model comparison used 98 compatible partitions; LASSO and elastic net had 95 valid evaluations because three additional single-predictor partitions were

unsupported by their fitting interface. Performance distributions were summarized by medians and interquartile ranges across valid outer test sets. These summaries describe variation across overlapping partitions. Execution denominators and exclusions are detailed in Table S5A.

Development-fitted models were applied to the fixed internal and external cohorts without refitting, retuning, or recalibration. The primary model was selected by joint consideration of AUROC, AP, Brier score, CITL, calibration slope, and consistency across the internal and external evaluations, without a prespecified weighted score or a single-metric ranking rule. External performance therefore contributed to model selection. The selected model was used for clinical benchmarking, risk stratification, decision-curve analysis, and prediction attribution.

### Clinical benchmarking and risk assessment

Comparators included logistic regression using Core4 and a single-predictor logistic benchmark using continuous MELD-Na. The MELD-Na benchmark was fitted in the development cohort and applied unchanged to the fixed evaluation cohorts. For the nested RBF-SVM–MELD-Na comparison, both models were fitted within the corresponding outer training partitions. Predictions from 98 paired outer test partitions were averaged for each patient using only occasions on which that patient was held out. Paired comparisons of AUROC, AP, and Brier score used these patient-level aggregates (Table S5D). This aggregate is an internal summary of repeated held-out predictions, rather than an additional independent validation cohort.

Low-, intermediate-, and high-risk strata were defined by the lower and upper tertiles of development-cohort RBF-SVM predictions and applied unchanged to the fixed internal and external cohorts. The cutoffs were approximately 0.234 and 0.559, calculated using linear interpolation (type 7 quantiles). We evaluated stratum-specific observed event rates, selected-observation coverage, event capture, and missed events. Calibration was assessed using CITL, calibration slope, and diagnostic calibration curves [13]. Decision-curve analysis (DCA) compared model net benefit across threshold probabilities [14]. Exploratory matched-capacity analyses compared observed-event capture at nominal selection proportions of 10%, 20%, 30%, 40%, and 50%. Exact stratum boundaries, calibration displays, and uncertainty methods are detailed in Note M4.

Exploratory sensitivity analyses examined alternative predictor representations, six Core4 ablations, and discrimination by event timing. For the timing analysis, the existing 1-year model scores were evaluated separately for events occurring within 90 days and after 90 days through day 365, each against the same 1-year non-event controls. These analyses did not fit new time-specific models (Tables S6 and S7).

### Prediction attribution

SHapley Additive exPlanations (SHAP) attributed the primary model's predictions to its input features [15]. Exact Kernel SHAP (kernelshap version 0.9.1) quantified contributions on the positive-class probability scale, using all 312 locked development records as the common empirical reference background. All feature coalitions were evaluated without sampling approximation. Global importance was summarized by mean absolute SHAP values; dependence plots related feature values to their contributions. Note M5 documents the background matrix and attribution checks.

Local SHAP decompositions were illustrated using two internal-validation cases with nearly identical MELD-Na scores but different RBF-SVM predictions. The cases were selected after inspection of predictions and outcomes. SHAP values describe attribution within the fitted model, and their interpretation accounts for the shared information between PLT and FIB-4.

### Statistical analysis

Continuous variables are summarized as medians and interquartile ranges and were compared with Mann–Whitney U tests. Categorical variables are presented as counts and percentages and were compared with chi-squared or Fisher exact tests. Standardized mean differences (SMDs) described case-mix differences between the Chinese and external cohorts.

Confidence intervals (CIs) for fixed-model performance, paired contrasts, and supplementary decision measures used 5,000 bootstrap resamples. Patients were resampled internally, whereas patient clusters were resampled externally, retaining all eligible hospitalization episodes and the multiplicity of each sampled patient. External point estimates were episode-weighted; clustering accounted for within-patient dependence in their uncertainty. Models shared resamples for paired contrasts, and 95% CIs were calculated from the 2.5th and 97.5th percentiles of valid bootstrap estimates. Risk-stratum event-rate intervals used Wilson intervals internally and patient-cluster bootstrap intervals externally. Calibration-group error bars used bootstrap intervals in both cohorts. Intervals based on fixed or aggregated predictions were conditional on those predictions and did not repeat model development.

All tests were two-sided, with $P < 0.05$ as the nominal significance threshold. Exploratory baseline comparisons, ablations, external model contrasts, and threshold-specific decision analyses were not adjusted for multiplicity. Analyses were performed primarily in R version 4.5.1 and Python version 3.12.14.

## Note M7 Extended results

### Cohort characteristics

Of 2,538 patients initially identified with HBV-related cirrhosis, 2,093 were excluded during eligibility screening and data curation, leaving 445 in the final analysis: 312 in development and 133 in fixed internal validation (Fig. 1). The primary outcome occurred in 185 patients (41.6%), including 130 development and 55 internal-validation patients. Overall characteristics are reported in Table S8, with outcome-stratified comparisons in Table S1. The external cohort comprised 102 eligible hospitalization episodes from 50 patients, including 67 event episodes (65.7%; Table S2).

In both Chinese subsets, patients subsequently readmitted had lower albumin and platelet counts and higher total bilirubin and INR; MELD-Na, ALBI, and FIB-4 also differed between outcome groups. Differences in sex, hepatitis B surface antigen levels, hepatitis B e antigen/antibody categories, ascites, variceal bleeding, spontaneous bacterial peritonitis, and infection were not statistically significant in either subset. HBV DNA levels and categories differed by outcome in development, but this pattern was not reproduced in fixed internal validation. Complete exploratory comparisons are provided in Table S1.

The external cohort differed from the Chinese cohort in age, INR, creatinine, serum sodium, FIB-4, and MELD-Na, whereas albumin, platelet count, and ALBI were more similar. Median FIB-4 and MELD-Na were 9.65 and 18.43 externally, compared with 5.76 and 12.47 in the Chinese cohort. Full characteristics and SMDs are reported in Table S2.

### Complementary selection identified four routine measures

LASSO retained five variables at λ1se, and Boruta retained ten after TentativeRoughFix (Fig. 2). Their intersection comprised INR, PLT, ALBI, and FIB-4. The categorical HBV DNA term was selected by LASSO but not Boruta and was therefore absent from Core4.

The largest absolute Spearman correlation among Core4 measures was 0.74, and VIFs ranged from 1.30 to 1.49, indicating limited residual linear redundancy. Across 100 nested selections, the intersection retained INR in 100%, ALBI in 85%, PLT in 67%, and FIB-4 in 46% of partitions. Median Jaccard similarity to the original Core4 set was 0.75 (interquartile range, 0.50–1.00), indicating that selection stability differed across predictors (Table S3B and C).

The tested alternative predictor representations and Core4 ablations did not identify a replacement or reduced set with consistently better performance across both evaluation cohorts. The raw-variable representation had similar internal point estimates but poorer external performance, and the composite representation showed no consistent advantage. Ablation effects varied by cohort and metric. These exploratory comparisons support retaining Core4 for the present model but do not establish that each measure contributes an independent effect or that reduced models are equivalent (Table S6).

### Internal performance was similar across candidate algorithms

Candidate models had broadly similar performance in fixed internal validation (Table 1; Fig. 3a and b). LASSO had AUROC, AP, and Brier score point estimates of 0.766, 0.721, and 0.192, respectively; corresponding values were 0.751, 0.672, and 0.199 for RBF-SVM and 0.747, 0.665, and 0.201 for LightGBM. RBF-SVM had the CITL closest to zero (0.020), although it did not lead the discrimination or overall-error comparisons.

Performance distributions overlapped in repeated nested internal evaluation. Median AUROC was approximately 0.78 for logistic regression, LASSO, elastic net, and RBF-SVM and was slightly lower for random forest, XGBoost, and LightGBM. AP and Brier score showed similar patterns. No algorithm consistently led the internal comparisons across metrics (Fig. 3c and d; Table S5A).

External evaluation differentiated the candidate models more clearly by point estimates. RBF-SVM achieved an AUROC of 0.770 (95% CI, 0.640–0.871), AP of 0.853, Brier score of 0.199, and calibration slope of 0.853. Random forest and XGBoost also had high AP estimates (0.834 and 0.849), but neither had better joint AUROC, Brier score, and calibration slope estimates than RBF-SVM. LightGBM had an AUROC of 0.694, AP of 0.799, and Brier score of 0.225. Across 5,000 patient-cluster bootstrap samples, RBF-SVM had the best mean rank for AUROC, AP, and Brier score. These exploratory rankings and contrasts describe the same fixed external predictions and do not establish uniform superiority (Table S5E and F).

RBF-SVM was selected for subsequent analyses because it combined similar internal performance with favorable external discrimination, probability error, and calibration slope point estimates (Fig. 3; Table 1). This selection reflected joint assessment of multiple metrics and cohorts; the external cohort consequently served both evaluation and model-selection roles.

### Comparison with the MELD-Na benchmark

In fixed internal validation, point estimates favored RBF-SVM over MELD-Na: the paired AUROC difference was +0.060 (95% CI, −0.021 to 0.141), AP difference was +0.052 (95% CI, −0.050 to 0.159), and Brier score improvement was +0.019 (95% CI, −0.011 to 0.048) (Fig. 4a; Table S5C).

External differences were +0.100 for AUROC (95% CI, −0.029 to 0.233), +0.122 for AP (95% CI, 0.001 to 0.230), and +0.015 for Brier score improvement (95% CI, −0.036 to 0.072). The AP interval excluded zero, while the AUROC and Brier intervals remained compatible with no difference. Overall, the estimated increments favored RBF-SVM but were imprecise.

Patient-level aggregates of paired held-out predictions from the repeated nested evaluation also favored RBF-SVM: AUROC difference, +0.061 (95% CI, 0.025–0.097); AP difference, +0.061 (95% CI, 0.012–0.116); and Brier score improvement, +0.024 (95% CI, 0.010–0.036). These comparisons provide internal support for the modeling procedure and are conditional on the aggregated predictions (Fig. 4a; Table S5D).

### Risk ordering persisted despite external underprediction

RBF-SVM predicted higher risks for observations followed by a target readmission in both evaluation cohorts (Fig. 4b). Applying the development-derived thresholds unchanged, observed event rates increased from low to intermediate to high risk: 15.9%, 40.0%, and 71.8% internally and 38.1%, 48.3%, and 86.5% externally. Risk ordering was preserved, but absolute event rates were higher in every external stratum (Fig. 4c).

The high-risk stratum covered 29.3% of internal-validation patients and captured 50.9% of their events; corresponding external values were 51.0% of episodes and 67.2% of events. Extending selection to the intermediate and high strata increased event capture to 87.3% internally and 88.1% externally, while covering 66.9% of patients and 79.4% of episodes, respectively (Table S9). These strategies quantify the trade-off between broader coverage and fewer missed observed events.

Internal RBF-SVM calibration estimates were a CITL of 0.020 and slope of 0.822. Externally, CITL was 0.725 (95% CI, 0.159–1.272) and slope was 0.853 (95% CI, 0.415–1.491), indicating overall underestimation of risk despite retained risk ordering (Fig. 4d and e; Table S5B).

Decision curves showed positive estimated RBF-SVM net benefit over several displayed threshold ranges in both cohorts (Fig. 4f and g). Matched-capacity analyses also showed enrichment of observed events among higher-ranked observations. Coverage, event capture, full-range decision curves, and paired net-benefit differences are reported in Figs. S1 and S2. These exploratory analyses concern hypothetical risk-triggered actions; they do not measure readmissions prevented through model-guided care.

### Discrimination by readmission timing

Discrimination was numerically higher for earlier than later readmissions. In internal validation, 22 events occurred within 90 days and 33 after 90 days through 1 year; AUROCs were 0.818 and 0.706, respectively,

with a later-minus-earlier difference of −0.112 (95% CI, −0.235 to 0.012). Externally, the corresponding 48 and 17 event episodes yielded AUROCs of 0.786 and 0.723 and a difference of −0.064 (95% CI, −0.266 to 0.077). Both timing comparisons used the same cohort-specific 1-year non-event controls, and both difference intervals included zero (Table S7).

### SHAP identified consistent model attribution across cohorts

The global attribution ranking was the same in both cohorts: INR, FIB-4, ALBI, and PLT. Mean absolute SHAP values were 0.108, 0.072, 0.056, and 0.032 in fixed internal validation and 0.112, 0.088, 0.059, and 0.039 externally (Fig. 5a and b).

Dependence plots showed nonlinear relationships between Core4 values and their contributions to model output. Higher INR, FIB-4, and ALBI generally contributed to higher predicted probabilities, whereas higher platelet counts generally contributed to lower probabilities (Fig. 5c–f).

Two illustrative internal-validation patients had nearly identical MELD-Na scores but different RBF-SVM predictions. The patient with a target readmission had MELD-Na 16.00 and a predicted probability of 0.803; the patient without an event had MELD-Na 15.99 and a probability of 0.221. Their INR values were 1.66 and 1.21, platelet counts were 36 and 92 × $10^9$/L, and FIB-4 values were 12.53 and 3.23, respectively. Local decompositions began from the same SHAP baseline and attributed the divergent predictions to different feature contributions (Fig. 5g and h).